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FACULTY OF SCIENCE

Integral Affine Structures

Classification of Integral Affine Structures on Compact Surfaces

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Preface

TODO: Add Preface

Summary

TODO: Add summary

List of symbols

$C^\infty(M, N)$	Smooth maps from M to N

Contents

CHAPTER ZERO

Preliminaries

0.1 Geometric Structures on Manifolds

0.1.1 Basic definition and properties

A lot of the content of this project extends in a nice way to more general types of geometries, namely locally homogeneous geometric structures. The idea behind these structures is that we can locally describe our "geometry" on a manifold M by charts into a space X together with a group G that tells us how to transition between charts on the overlaps.

We will require that our group G be a Lie Group acting transitively on M.

Definition 0.1. Let $U \subset X$ be an open set of our model space X. We say that a smooth map $f : U \mapsto X$ is *locally*-G if for each component $U_i \subset U$ there is a $g_i \in G$ such that $f|_{U_i} = g_i|_{U_i}$

Remark 0.2. Locally-G maps satisfy the following unique extension property. If $U \subset X$ and $f : U \mapsto X$ is a locally-G map then there is a unique $g \in G$ restricting to f.

Definition 0.3. A (G, X) atlas is a pair $(\mathcal{U}, \Phi)$ consisting of an open covering for M and a collection of charts such that the transition maps are locally-G on their respective overlaps.

Definition 0.4. A map $f : M \mapsto N$ between two (G, X)-manifolds is a (G,X)-map if for each pair of charts (ϕ, φ) then $\phi \circ f \circ \varphi^{-1}$ is locally-G on the relevant intersection.

TODO: Add some examples (euclidean/projective/hyperbolic) TODO: At some point include classification of 1-dim integral affine (This is easy in - textbook)

0.1.2 The developing map and holonomy representation

For (G,X)-manifolds, there is an inherently nice way to "globalize" the charts via a (G,X)-map called the developing map. The study of this map will be of importance to us throughout this thesis.

Proposition 0.5. Let M be a simply connected (G, X)-manifold. Then there exists a (G, X)-map

$$\mathrm{dev} : M \mapsto X$$

Proof. We pick a base point $x_0 \in M$ and a chart (U_0, φ_0) around x_0. Now for any $x \in M$ we define the map dev as follows. We pick a curve $x(t)$ on M satisfying $x(0) = x_0$ and $x(1) = x$. As the image of our curve is compact, we can cover the it by finitely many coordinate charts U_i with $i \in \{0, \ldots, n\}$ such that $x(t) \in U_i$ for $t \in (a_i, b_i)$ with

$$a_0 < 0 < a_1 < b_0 < a_2 < b_1 < a_3 \cdots < a_n < b_{n-1} < 1 < b_n$$

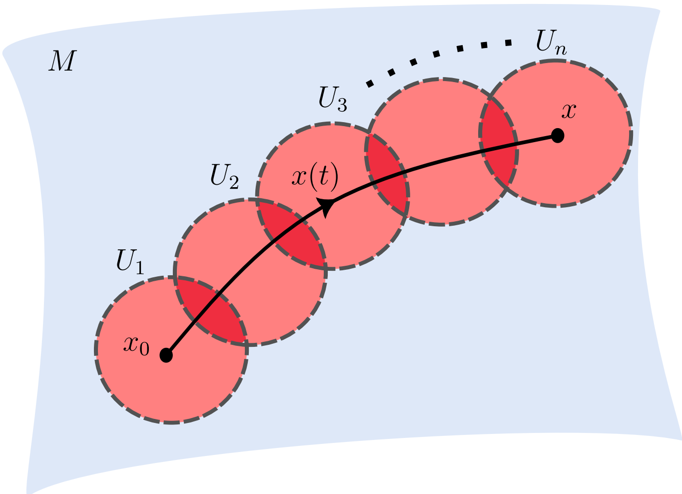

Figure 1: Developing along a curve

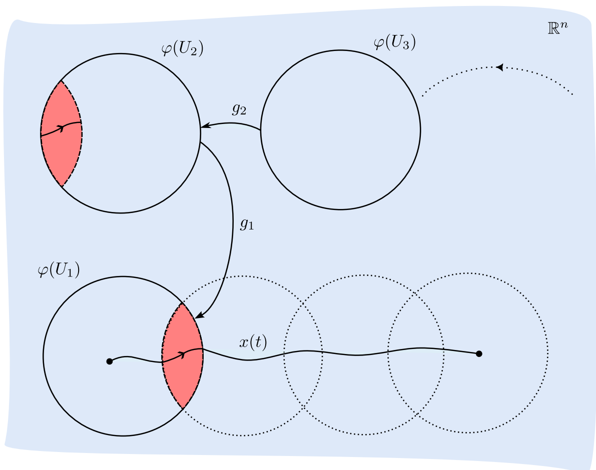

Figure 2: We successively glue our charts together using the transition maps

Now we consider where these charts overlap. Since our manifold is equipped with a (G, X)-structure our transition maps differ by an element of G. Let $g_i = \varphi_{i-1} \circ \varphi_i^{-1} \in G$ be the transition map from U_i and U_{i-1}. We then define

$$\text{dev}(x) = g_1 g_2 \dots g_n \varphi_n(x)$$

We must show that this is a well-defined map. We first show that this map does not change if we refine the cover. Suppose we insert a chart (V, ϕ) between U_i and U_{i-1}. Let $h_i, h_{i-1} \in G$ be the unique group elements such that

$$\varphi_{i-1} = h_{i-1} \circ \phi \text{ on } V \cap U_{i-1} \text{ and}$$
$$\phi = h_i \varphi_i \text{ on } V \cap U_{i-1}.$$

Then on $V \cap U_i \cap U_{i-1}$ we have that $\varphi_{i-1} = h_{i-1} \circ h_i \circ \varphi_i$. This gives us that $g_i = \varphi_{i-1} \circ \varphi_i^{-1} = h_{i-1} \circ h_i$ as we require that our transition maps satisfy a unique extension property.

Now when we develop along our curve with respect to this new covering, we obtain that:

$$\text{dev}(x) = g_1 \dots g_{i-1} h_{i-1} h_i g_{i+1} \dots g_n \varphi_n(x) = g_1 \dots g_{i-1} g_i g_{i+1} \dots g_n \varphi_n(x)$$

This defines dev along $x(t)$. We now need to show that it does not depend on our choice of curve. As M is simply connected, all curves with the same start point and end point will be homotopic. Indeed, by compactness we can split our homotopy into smaller homotopies such that we can break up our curve into regions

$$a_1 = 0 < a_2 < \dots < a_n = 1$$

such that during each small homotopy the segment $x((a_i, a_{i+1}))$ lies entirely inside a coordinate patch. It then follows that dev is independent of our choice of paths completing the proof.

□

This shows that we have a well defined development map on any simply connected (G, X)-manifold. However, this map depended on the choice of base point and initial chart. We immediately get the following corollary from this fact.

Corollary 0.6. Let M be a (G, X)-manifold and let $\tilde{M}$ be its universal cover. Then there exists a pair (dev, hol) satisfying

$$\text{dev} : \tilde{M} \mapsto X$$
$$\text{hol} : \pi_1(M) \mapsto G$$

where hol(γ) is the element $g_1 \dots g_n$ of G obtained from developing around a loop $\gamma \in M$.

The developing pair satisfy the following equivariance condition. Let $\gamma \in \pi_1(M)$ act by deck transformations on $\tilde{M}$. Then we have the following. (TODO: Fix this proof)

Lemma 0.7. Let $\gamma \in \pi_1(M)$. Then

$$\text{dev}(\gamma \cdot x) = \text{hol}(\gamma)\text{dev}(x)$$

for $x \in \tilde{M}$.

Proof. Let $\{(V_i, \phi_i)\}$ be a cover for γ and lift this to a cover of a lift of γ which we denote by $\tilde{\gamma}$. Denote the transition maps on $V_i \cap V_{i+1}$ by g_i. Then by definition

$$\begin{aligned} \mathrm{dev}(\gamma \cdot x) &= g_1 \dots g_n \varphi_k(\gamma(1)) \\ &= \mathrm{hol}(\gamma)\mathrm{dev}(x) \end{aligned}$$

□

The role of the development pair is to "globalize" the coordinate chart of M. Indeed the following proposition makes this statement more clear.

Proposition 0.8. Let M be a (G, X)-manifold with development pair (dev, hol). Denote by Γ the image of $\pi_1(M)$ under the holonomy map. Then there exists a (G, X)-atlas such that the transition maps lie in Γ.

Proof. We pick $m \in M$ and fix a lift $\tilde{m} \in \tilde{M}$ of m. Then, as dev is a local diffeomorphism, there is an open set $\tilde{U}$ of $\tilde{m}$ such that dev is a local diffeomorphism onto X.

We pick $V \subset M$ an open neighborhood of m and take its inverse under the covering map $p^{-1}(V)$ which we may assume dev is a diffeomorphism on by intersetcting with $\tilde{U}$.

(TODO: Make this proof more clear) Now, define coordinate charts on M by $(V_\alpha, \psi_\alpha = \mathrm{dev} \circ p_\alpha^{-1})$ at a point $m \in M$ where by p_α^{-1} we mean we have fixed a component of the preimage p^{-1} restricted to V_α. Now suppose that $V_i \cap V_j \neq \emptyset$ and let $\tilde{U}_i = p_i^{-1}(V_i)$ and $\tilde{U}_j = p_j^{-1}(V_j)$. Then the preimage of their intersection will differ by a deck transformation $\gamma \in \pi_1(M)$. Therefore, on the intersection $V_i \cap V_j$

$$\begin{aligned} \mathrm{dev} \circ p_i^{-1} &= \mathrm{dev} \circ \gamma \circ p_j^{-1} \\ \Rightarrow p_i^{-1} \circ p_j &= \gamma \end{aligned}$$

Now our by definition of our transition maps

$$\begin{aligned} \psi_i \circ \psi_j^{-1} &= \mathrm{dev} \circ p_{i-1} \circ p_j \circ dev^{-1} \\ = \mathrm{dev} \circ \gamma \circ dev^{-1} &= \mathrm{hol}(\gamma) dev \circ dev^{-1} = \mathrm{hol}(\gamma) \in \Gamma \end{aligned}$$

and so we have thus given our coordinate charts on M in terms of the development pair. □

This shows that once we have fixed a development pair, our (G, X)-structure is entirely determined by this. Conversely, the following lemma shows us that once we have fixed our (G, X)-structure, the development pair is determined up to certain factors/conjugations.

Lemma 0.9. Let M be a (G, X)-manifold with development pair $(\mathrm{dev}, \mathrm{hol})$. Suppose that $(\mathrm{dev}', \mathrm{hol}')$ is another development pair for M. Then $\mathrm{dev}' = g \circ \mathrm{dev}$ and $\mathrm{hol}' = g \circ \mathrm{hol} \circ g^{-1}$ for some $g \in G$.

Proof. Our development pair was determined once we picked a base point and a chart (U, ϕ). Suppose that we pick a different base point and chart (V, φ). Now we consider how we define $\mathrm{dev}(x)$ for $x \in \tilde{M}$. We choose a curve γ connecting x_0 and x and choose a cover U_i of γ and develop around it. Now, with respect to our new initial chart, $V \cup \{U_i\}$ is an open cover of γ. If we develop starting with this new chart, we can first glue all the U_i to U_1 as before and then glue this chain to V with an element of G by gluing U_1 to V on their overlap. Hence the entire development will have just changed by that element of G.

If, on the other hand, both the base point and chart have changed, we can pick a curve γ' starting at our new base point x_0' and ending at our old basepoint x_0. Developing first along this curve and then along our old curve γ we pick up some element of G from developing γ' whilst developing along γ gives the same as before. As we already showed this map is homotopy invariant this gives that $dev'(x) = g \circ \text{dev}(x)$ for some $g \in G$.

Now to show that $hol' = g \circ \text{hol} \circ g^{-1}$ we consider

$$\begin{aligned} dev'(\gamma \star \alpha) &= g \circ \text{dev}(\gamma \star \alpha) \\ &= g \circ \text{hol}(\gamma) \circ \text{dev}(\alpha) \\ &= (g \circ \text{hol}(\gamma) \circ g^{-1})(g \circ \text{dev}(\alpha)) \\ &= (g \circ \text{hol}(\gamma) \circ g^{-1})\text{dev}'(\alpha) \end{aligned}$$

and as the development pair satisfied the equivariance property it follows that $hol' = g \circ \text{hol} \circ g^{-1}$. □

This shows that once we have fixed a development pair, any other development pair can is related by $(\text{dev}', \text{hol}') = (g \circ \text{dev}, g \circ \text{hol} \circ g^{-1})$. We should therefore note that if we want to classify (G, X)-structures on a manifold M by classifying the possible development pairs, it is important to consider them up to this relation.

0.2 Completeness of (G,X)-structures

An important property the developing map can have is that of *completeness*.

Definition 0.10. A (G, X)-manifold M is said to be complete if its developing map is a diffeomorphism onto X.

In the case that X is simply connected e.g. $X = \mathbb{R}^n$, we have that M is diffeomorphic to X/Γ where $\Gamma = \text{hol}(\pi_1(M))$.

In the case that M is a Riemannian manifold, this notion of completeness is equivalent to completeness of the Levi-Civita connection of M.

TODO : Examples of complete/ incomplete manifolds e.g. Hopf

0.2.1 Affine Structures on the Torus

The standard affine structure on the torus is obtained from viewing the torus as $\mathbb{R}^n/\mathbb{Z}^n$ where $\mathbb{Z}^n$ acts by translations. This structure is easily seen to be a complete integral affine manifold. Indeed, the action is an action by the integral affine group and is both free and proper. Covering space theory tells us that $\pi : \mathbb{R}^n \to \mathbb{R}^n/\mathbb{Z}^n$ is a covering space with the group of deck transformations via our $\mathbb{Z}^n$ action. A theorem of Auslander and Markus (L. Auslander and L. Markus, Holonomy of flat affinely connected manifolds) tells us that affine structures obtained in this manner are complete. However, even for the 2-torus, not all affine structures are complete. We now give an example of an incomplete structure on $\mathbb{T}^2$.

Example 0.11. Let $\lambda > 1$ be a real number and consider the action of λ on $\mathbb{R}^2 - \{0\}$ given by multiplication. This action is a free and proper action by the affine group and so the quotient $M = \mathbb{R}^2 - \{0\}/\langle\lambda\rangle$ is an affine manifold. The map

$$\rho : \mathbb{R}^2 \to \mathbb{R}^2 - \{0\}$$
$$(x, y) \mapsto e^x(\cos y, \sin y)$$

is a covering map. Basing our fundamental group at $[(1, 0)] \in M$, our group is generated by the loops γ_1 and γ_2 where γ_1 is the projection of the line segment joining the point $(1, 0) \mapsto (\lambda, 0)$ and γ_2 is the loop going around the unit circle once. These lift to curves $\bar{\gamma}_1$ and $\bar{\gamma}_2$ in the universal cover satisfying

$$\bar{\gamma}_1(0) = (0, ln(1)) = (0, 0), \quad \bar{\gamma}_1(1) = (0, ln(\lambda)),$$
$$\bar{\gamma}_2(0) = (0, 0), \quad \bar{\gamma}_2(1) = (0, 2\pi),$$

from this we see that γ_1 and γ_2 act on (x, y) by

$$\gamma_1 \cdot (x, y) = (ln(\gamma) + x, y)$$
$$\gamma_2(x, y) = (x, y + 2\pi)$$

Now, we define our developing pair by

$$\mathrm{dev}(x, y) = \rho(x, y),$$
$$\mathrm{hol}(\gamma_1) \cdot (x, y) = (ln(\lambda) + x, y),$$
$$\mathrm{hol}(\gamma_1) = id_{\mathbb{R}^2}$$

We check that these satisfy the equivariant property.

$$\begin{aligned}\mathrm{dev}(\gamma_1 \cdot (x, y)) &= \mathrm{dev}((\ln(\lambda) + x, y))\\ &= e^{ln(\lambda)} e^x(\cos y, \sin y)\\ &= \lambda dev((x, y))\\ &= hol(\gamma_1)dev((x, y))\end{aligned}$$

$$\begin{aligned}\mathrm{dev}(\gamma_2 \cdot (x, y)) &= \mathrm{dev}((x, y + 2\pi))\\ &= e^x(\cos(y + 2\pi), \sin(y + 2\pi))\\ &= e^x(\cos y, \sin y)\\ &= hol(\gamma_2)dev((x, y))\end{aligned}$$

It is now clear that this structure is incomplete as $0 \notin im(\mathrm{dev}(\mathbb{R}^2))$. Indeed, it is typical that the fixed point of a radiant structure is not in the developing image as shown in (TODO: Theorem 6.5.2 Geometric Structures)

0.3 Representation Theory and Affine Manifolds

0.3.1 Affine representations

We now turn our attention to affine manifolds before specializing to the integral affine case. An affine manifold is a (G, X)-manifold where $X = \mathbb{R}^n$ and $G = \mathrm{Aff}(\mathbb{R}^n)$ An important tool in classifying affine structures is the representation theory of groups. Often, by studying properties of the holonomy group of an affine manifold M, we are able to obtain information about the affine structure of M. Of fundamental importance is the case where the holonomy group is nilpotent which we will shortly see. We will use these tools to determine conditions for the affine structure to be complete.

Definition 0.12. An affine transformation of $\mathbb{R}^n$ is a map

$$A : \mathbb{R}^n \mapsto \mathbb{R}^n$$

such that $A(x) = Gx + b$ where $G \in GL(\mathbb{R}^n)$ and $b \in \mathbb{R}^n$

The group $\text{Aff}(\mathbb{R}^n)$ is the group of affine transformations of $\mathbb{R}^n$ and is equal to $\text{Aff}(\mathbb{R}^n) = \mathbb{R}^n \rtimes GL(\mathbb{R}^n)$

Definition 0.13. An affine representation of a group G is a group homomorphism

$$\alpha : G \mapsto \text{Aff}(\mathbb{R}^n)$$

Example 0.14. Let M be a (G, X)-manifold with $G = \text{Aff}(\mathbb{R}^n)$ and $X = -\mathbb{R}^n$. Then the holonomy representation is an affine representation of $\pi_1(M)$.

We have that an affine representation splits into a linear part and a translation, e.g. $\alpha(g) = (\lambda(g), \mu(g))$. The linear part makes $\mathbb{R}^n$ into a G-module with the obvious action which we will denote E.

We will be interested in some results from group cohomology. This leads us to the following definitions.

Definition 0.15 (Crossed Homomorphism)**.** A crossed homomorphism for λ, is a group homomorphism

$$\mu : G \mapsto E$$

that satisfies $\mu(gh) = \mu(g) + \lambda(g)\mu(h)$

Proposition 0.16. Let $\alpha(g) = (\lambda(g), \mu)$ be an affine representation on E of a group G. Then μ is a crossed homomorphism for λ

Proof. Let $g, h \in G$ and $\alpha = \mu + \lambda$ be an affine representation. Then

$$\begin{aligned} \alpha(gh) &= (\mu(gh), \lambda(gh)) \\ &= \alpha(g) \cdot \alpha(h) = (\mu(g), \lambda(g)) \cdot (\mu(h), \lambda(h)) \\ &= (\mu(g) + \lambda(g)\mu(h), \lambda(g)\lambda(h)) \\ &\Rightarrow \mu(gh) = \mu(g) + \lambda(g)\mu(h) \end{aligned}$$

and so μ is a crossed homomorphism for λ. $\square$

Definition 0.17 (Principle Crossed Homomorphism)**.** A *principle crossed homomorphism* for λ is a crossed homomorphism of the form $\mu(g) = y - \lambda(g)y$ for some $y \in E$.

0.3.2 Radiant affine manifolds

Of importance to completeness of the affine structure is the case that our affine representation has a fixed point. It turns out that such structures will always be incomplete.

Definition 0.18. An affine representation with a fixed point is called *radiant*.

Proposition 0.19. Let $y \in E$. Then y is a *stationary point* of the action of α if and only if $\mu(g) = y - \lambda(g)y$ for all $g \in G$. TODO(why 2 labels?)

Proof. Suppose $y \in E$ is a ary point of α. Then $\alpha(g)y = y$ for every $g \in G$. This gives that

$$(\lambda(g) + \mu(g))y = \lambda(g)y + \mu(g) = y$$
$$\Rightarrow \mu(g) = y - \lambda(g)y$$

Conversely, suppose that $\mu(g) = y - \lambda(g)y$. Then $y = \mu(g) + \lambda(g)y = \alpha(g)y$ □

Remark 0.20. If μ is a principle crossed homomorphism we will often write $\mu = D_y$

TODO: Develop some of the basics of group cohomology

Lemma 0.21. The cohomology group $H^1(G, E)$ is isomorphic to

$$H^1(G, E) \cong \frac{\{\text{crossed homomorphisms for } \lambda\}}{\{\text{principle crossed homomorphisms for } \lambda\}}$$

Proof. **TODO : Add proof e.g. construct resolution or take this as given?** □

Definition 0.22 (Radiance Obstruction)**.** Let α be an affine representation. The radiance obstruction of α is the cohomology class

$$c_\alpha = [\mu] \in H^1(G, E)$$

where μ is the translational part of α.

Lemma 0.23. The radiance obstruction vanishes, e.g. $c_\alpha = 0$ if and only if α is conjugate to its linear part by a translation, if and only if α has a stationary point.

Proof. By Lemma 0.19, $y \in E$ is stationary if and only if $\mu(g) = D_y$ is a principle crossed homomorphism for λ if and only if $c_\alpha = [\mu] = 0$.

Now suppose that $\mu = D_y$ for some $y \in E$. so that $\mu(g) = y - \lambda(g)y$. We define the translation

$$T_y : E \mapsto E$$
$$x \mapsto x - y$$

Now, conjugating α by this translation yields

$$T_y \circ \alpha(g) \circ T_y^{-1} = T_y \circ \alpha(g)(x + y)$$
$$= T_y \circ (\lambda(g)x + \lambda(g)y + \mu(g))$$
$$= T_y \circ (\lambda(g)x + \lambda(g)y + y - \lambda(g)y)$$
$$= T_y \circ (\lambda(g)x + y) = \lambda(g)x + y - y = \lambda(g)x$$

Conversely, suppose that $T_y \circ \alpha(g) \circ T_y^{-1} = \lambda(g)$ for all $g \in G$ and $x \in E$. Then

$$\begin{aligned} &T_y \circ \alpha(g) \circ T_y(x) = T_y \circ \alpha(g)(x+y) \\ &= T_y \circ (\lambda(g)(x+y) + \mu(g)) = \lambda(g)(x+y) + \mu(g) - y = \lambda(g)x \\ &\Rightarrow \mu(g) = \lambda(g)(x) - \lambda(g)(x) - \lambda(g)(y) + y \\ &= y - \lambda(g)y \end{aligned}$$

as required. □

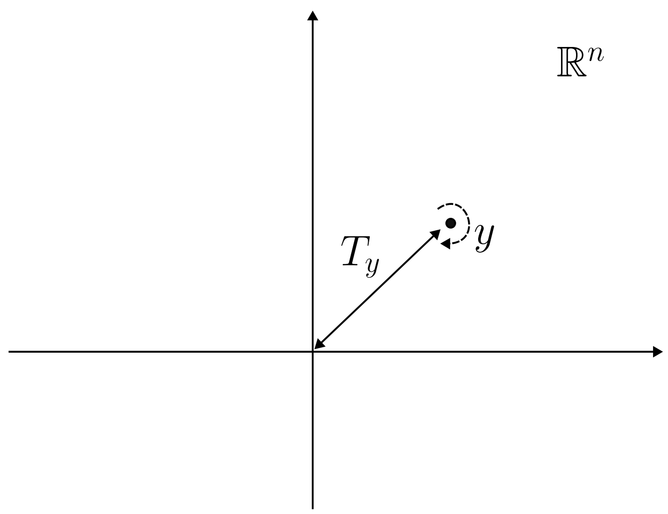

Figure 3: Conjugating a fixed point y under the affine action by a translation to the origin

Remark 0.24. This lemma will allow us to alter the development map by a translation (which alters the holonomy representation by a conjugation) to assume that our fixed point is at the origin.

Definition 0.25 (Contragradient Representation)**.** Let E be a G-module and α a representation on E. We define the contragradient representation on E^* by

$$\begin{aligned} \alpha^* : G &\mapsto GL(E^*) \\ g &\mapsto \alpha^* g \end{aligned}$$

such that $\alpha^* g(f)(x) = f(\alpha(g^{-1})x)$ for all $x \in E$ and $f \in E^*$.

TODO - Fix a notation for these representations.

Remark 0.26. This induces a representation on $\wedge^k E^*$ by
$(g \cdot \omega)(x_1, \ldots, x_n) = \omega(g^{-1} \cdot x_1, \ldots, g^{-1} \cdot x_n)$ for $\omega \in \wedge^k E^*$ and $x_i \in E$.

We will make use of the following lemma when looking at the relationship between the holonomy representation and parallel volume forms.

Lemma 0.27. Let $F : V \mapsto V$ be a linear map from a vector space V of dimension n to itself and let $\omega \in \wedge^n(V^*)$. Then

$$F^*\omega = \det(A)\omega$$

where A is any matrix representing the linear transformation F.

Proof. As $dim(\wedge^n(V^*)) = 1$ and F is a linear map, it follows that our map corresponds to multiplication by some scalar $\lambda \in \mathbb{R}$. Let $\varphi : V \mapsto \mathbb{R}^n$ be an isomorphism identifying our vector space V with $\mathbb{R}$. Then we have

$$\begin{gathered} F^*\varphi^* \det = \lambda\varphi^* \det \\ \Rightarrow (\varphi^{-1})^* F^*\varphi^* \det = \lambda \det \\ \Rightarrow (\varphi^{-1} F\varphi)^* \det = \lambda \det \end{gathered}$$

If we let $A = \varphi^{-1}F\varphi$ and $\{e_1, \ldots, e_n\}$ the standard basis of $\mathbb{R}^n$ then

$$A^* \det(e_1 \ldots, e_n) = \lambda \det(e_1, \ldots, e_n) = \lambda = \det(Ae_1, \ldots, Ae_n)$$

which gives us that $\det(A) = \lambda$ as required. $\square$

This leads us to the following elementary but important property of the action of the holonomy representation on volume forms.

Corollary 0.28. Let M be an affine manifold and $\lambda \in GL(\mathbb{R}^n)$ be the linear part of the holonomy representation. Then M has a parallel volume form with respect to the affine connection if and only if $\det(\lambda(g)) = 1$ for all $g \in G$.

Proof. Suppose ω is a parallel volume form on M. Then the parallel transport of ω around a loop in M leaves ω unchanged, e.g. the linear holonomy action preserves ω. This gives us that

$$\begin{gathered} g \cdot \omega = \omega \\ \Rightarrow \omega(g^{-1} \cdot x_1, \ldots, g^{-1} \cdot x_n) = \omega(x_1, \ldots, x_n) \end{gathered}$$

for all $x_i \in M$. But by Lemma 0.27

$$\omega(g^{-1} \cdot x_1, \ldots, g^{-1} \cdot x_n) = \det(\lambda(g^{-1})) \cdot \omega(x_1, \ldots, x_n)$$

and so the volume is preserved if $\det(\lambda(g)) = 1$

Conversely, suppose that the determinant of the linear holonomy is 1 for all $g \in G$. If we pick a point $m_0 \in M$ then locally there exists a volume form ω_0 on a chart U of M with respect to the affine connection, corresponding to a constant, top degree form. We now claim that we can uniquely extend such a form. Indeed, as parallel transport provides a linear isomorphism between tangent spaces we can attempt to define a volume form on M by parallel transporting ω_0.

Pick another point $m \in M$. We claim that the parallel transport of ω_0 along a curve γ from m_0 to m does not depend on the choice of curve. If γ' is another curve then the parallel transport $(P^{\gamma'}_{m_0 \to m})^{-1} \circ P^{\gamma}_{m_0 \to m} = g$ for some g in the linear holonomy group. It follows that

$$\begin{gathered} g^* \circ \omega_0 = ((P^{\gamma'}_{m_0 \to m})^{-1} \circ P^{\gamma}_{m_0 \to m})^* \circ \omega_0 \\ = \det(\lambda(g)) \cdot \omega_0 = \omega_0 \\ \Rightarrow (P^{\gamma}_{m_0 \to m})^* \circ \omega_0 = (P^{\gamma'}_{m_0 \to m})^* \circ \omega_0 \end{gathered}$$

and so the parallel transport does from m to m_0 does not depend on the choice of path. We note that this defines a continuous maps as in each affine chart the parallel transport identifies the tangent spaces via the identity map and on over laps they differ by an element of $GL(\mathbb{R}^n)$ as the derivative of an affine map extracts the linear part. $\square$

Definition 0.29 (Radiant Manifold)**.** An affine manifold is said to be radiant if its affine holonomy representation has a fixed point.

Remark 0.30. By Lemma 0.23 we may conjugate our developing map by a translation to assume that the fixed point is at the origin.

When M is radiant the above assumption shows that the radiant vector field $R = \sum_i x_i \frac{\partial}{\partial x_i}$ is fixed by the affine action. There is then a unique vector field $\tilde{X}$ on the universal cover $\tilde{M}$ that is related to R by the developing map. Since this is preserved by deck transformations (by equivariance of developing map and since the representation is radiant) it descends to a vector field X on M.

Definition 0.31. The vector field X on M is called the radiant vector field of M.

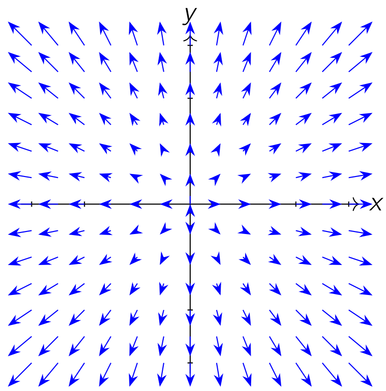

Figure 4: Radiant Vector Field on $\mathbb{R}^2$

0.4 Affine Manifolds and Integral Affine Manifolds

CHAPTER ONE

Nilpotent Holonomy and Completeness

1.1 Nilpotent Holonomy

1.1.1 Nilpotent Groups

In this section, we will see that the condition that the holonomy group of an affine manifold M be nilpotent will be useful in determining whether the affine structure is complete. We will build up some useful results before determining some equivalent conditions that force completeness of the affine structure.

Definition 1.1 (Central Series)**.** Let G be a group. Then a central series for G is a finite collection of normal subgroups such that

$$1 = G_0 \leq G_1 \leq \cdots \leq G_s = G$$

and

$$\frac{G_{i+1}}{G_i} \leq Z(\frac{G}{G_i})$$

where $Z(\frac{G}{G_i})$ denotes the center.

Definition 1.2 (Nilpotent Group)**.** A group G is said to be nilpotent if it admits a central series.

Remark 1.3. The *nilpotent class* of G is the length of the shortest central series

(TODO: Examples of nilpotent groups) For nilpotent groups, we have the following results regarding their group cohomology.

Lemma 1.4. Let G be a nilpotent group and E a G-module. If $H^0(G, E) = 0$ then also $H^i(G, E) = 0$ for all $i \geq 0$.

Proof. TODO : Add proof? - "HIRSCH, M., Flat manifolds and the cohomology of groups" □

from this we obtain the following lemma

Lemma 1.5. If G is a nilpotent group and E is a G-module, then the following statements are equivalent.

1. $H^0(G, E) = 0$
2. $H^0(G, E^*) = 0$
3. $H_0(G, E) = 0$
4. $H_0(G, E^*) = 0$

Proof. See [FGH81] □

1.1.2 Unipotence and the Fitting submodule

In the case that the affine holonomy group is nilpotent, we will be able to show that completeness is equivalent to a property called unipotence of the linear holonomy. We will now outline some consequences of unipotence.

Definition 1.6 (Unipotent)**.** A G-module E is called unipotent if $(g - I)^n = 0$ for every $g \in G$ where $n = dim(E)$.

Definition 1.7. An affine representation is unipotent if its linear part defines a unipotent G-module.

Although E may not be unipotent itself, we can consider the smallest submodule (possibly the zero module) of E which is unipotent. We shall denote this submodule by E_U.

Remark 1.8. The submodule E_U is called the *Fitting submodule* of E.

Often, it will be useful to quotient out by the unipotent part of a G-module. The following lemma gives some insight on why this is often a useful idea.

Lemma 1.9. Let E_U be the Fitting submodule of E. Then

$$H^0(G, E/E_U) = 0$$

Proof. Take E_U to be the Fitting submodule and assume $E_U \neq E$ otherwise the result is trivial. As $H^0(G, E/E_U)$ is the set of ary points of the G-module E/E_U, asuuming $H^0(G, E/E_U) \neq 0$ we can take $x + E_U \in E/E_U$ not the identity so that

$$g \cdot (x + E_U) = x + E_U \text{ for all } g \in G$$
$$\Rightarrow (g - 1) \cdot x \in E_U \text{ for all } x \in E$$

Now we consider the G-submodule spanned by x and E_U. As $x \notin E_U$ this submodule is strictly bigger than E_U. Now we have that

$$(g - 1)^n \cdot (x + e) = (g - 1)^{n-1}((g - 1)x + (g - 1)e)$$

But $(g-1)x \in E_u$ so that $(g-1)x+(g-1)e \in E_U$. But as $E_U \neq E$ it has dimension at most $n-1$ and so $(g-1)^{n-1}=0$ restricted to E_U. Therefore this is a unipotent submodule strictly bigger than E_U which contradicts maximality. □

The following splitting lemma will prove important in our characterization of completeness.

Lemma 1.10. If G is a nilpotent group and E a G-module with Fitting submodule E_U, then there exists a unique submodule F such that $E = E_U \oplus F$.

Proof. Let β be the induced representation on $GL(E/E_U)$. By Lemma 1.9, $H^0(G, E/E_U) = 0$.

Claim. A submodule $F \subset E$ is complementary to E_U e.g. $E = E_U \oplus F$ if and only if $F = T(E/E_U)$ where $T : E/E_U \mapsto E$ is an equivariant linear map with $P \circ T = 1_{E/E_U}$ with P the canonical projection map.

Now let $S : E/E_U \mapsto E$ be a linear map satisfying $P \circ S = 1_{E/E_U}$. Then any such T as above can be uniquely written as $T = R + S$ with $R : E/E_U \mapsto E$. It is T-equivariant if and only if

$$R + S = g \circ (R + S) \circ \beta(g)^{-1}$$

$$\Rightarrow R = g \circ R \circ \beta(g)^{-1} + g \circ S \circ \beta(g)^{-1} - S \tag{1.1}$$

If we can show that there is a unique such T, then we have shown that E splits uniquely into $E = E_U \oplus F$. This will follow if we can show there is a unique R satisfying the above.

To do this we define a representation on $Hom(E/E_U, E)$ by

$$\begin{gathered}\gamma : G \mapsto Hom(E/E_U, E)\\ \gamma(g)(R) = g \circ R \circ \beta(g)^{-1}\end{gathered}$$

and we define a crossed homomorphism for γ by $\mu(g) = g \circ S \circ \beta(g)^{-1} - S$. This defines a crossed homomorphism as

$$\begin{aligned}\mu(gh) &= g \circ h \circ S \circ \beta(h)^{-1} \circ \beta(g)^{-1} - S\\ &= \gamma(g) \circ h \circ S \circ \beta(h)^{-1} - S\\ &= \gamma(g) \circ (\mu(h) + S) - S\\ &= \gamma(g) \circ \mu(h) + g \circ S \circ \beta(g)^{-1} - S\\ &= \gamma(g)\mu(h) + \mu(g)\end{aligned}$$

As P is equivariant and $P \circ S = 1_{E/E_U}$ we get

$$\begin{gathered}P \circ \mu(g) = P \circ g \circ S \circ \beta(g)^{-1} - P \circ S, \text{ so by equivariance}\\ = \beta(g) \circ P \circ S \circ \beta(g)^{-1} - 1\\ \beta(g) \circ \beta(g)^{-1} - 1 = 0\end{gathered}$$

Now we can we write (1.1) as

$$R = \gamma(g) \circ R + \mu(g)$$

which by Proposition 0.19 is saying that R is a fixed point of the affine representation induced by γ and μ. We wish now to show that this affine representation has a unique fixed point. This is equivalent by Lemma 1.4 and Lemma 1.5 to $H^0(G, Hom(E/E_U)) = 0$.

To this end we suppose that $R : E/E_U \mapsto E$ is fixed under all $\gamma(g)$ for $g \in G$. Then, letting $d = dim(E_U)$ for any possibly the same collection of d elements $g_1, \ldots, g_d \in G$ we have

$$(I - g_1) \circ \cdots \circ (I - g_d)|E_U = 0$$

as they act unipotently on E_U. It now follows by equivariance that

$$\begin{gathered}(I - g_1) \circ \cdots \circ (I - g_d) \circ R \\ = (I - g_1) \circ \cdots \circ (R - g_d \circ R) \\ = (I - g_1) \circ \cdots \circ (R - R \circ \beta(g_d)) \\ = (I - g_1) \circ \cdots \circ R \circ (1 - \beta(g_d)) \\ \Rightarrow R \circ (1 - \beta(g_1)) \circ \cdots \circ (1 - \beta(g_d)) = 0\end{gathered}$$

so that R vanishes on vectors of the form

$$(1 - \beta(g_1)) \circ \cdots \circ (1 - \beta(g_d))x$$

for $x \in E/E_U$. Thus if all vectors are in this span, then the only such R fixed under the action of γ is the zero map. As $(1 - \beta(g))$ is invertible on E/E_U (it acts without unipotent part), this is equivalent to showing that E/E_U is spanned by vectors of the form $(1 - \beta(g))$. By definition, this is equivalent to $H_0(G, E/E_U) = 0$ which is true by 1.9 completing the proof. □

This proves our result once we have cleared up that the stated claim is indeed true which we show in the following proposition.

Proposition 1.11. A submodule $F \subset E$ is complementary to E_U e.g. $E = E_U \oplus F$ if and only if $F = T(E/E_U)$ where $T : E/E_U \mapsto E$ is an equivariant linear map with $P \circ T = 1_{E/E_U}$ with P the canonical projection map.

Proof. Suppose $F = T(E/E_U)$, with T being G-equivariant so that F is a G-module. If $T(x + E_U) = e \in E$, then $P(e) = e + E_U = x + E_U \Rightarrow x \in E_U$ and so by linearity $e = 0$ and so $F \cap E_U = \{0\}$. Clearly, $E_U \oplus F \subseteq E$. For any $x \in E$, $x - T(x + E_U) \in E_U$. We can thus write $x = (x - T(x + E_U)) + T(x + E_U) \in E_U + F$.

Conversely, suppose that F is a G-submodule and $F \cap E_U = \{0\}$ and $F \oplus E_U = E$. Then its easy to see that $P|_F : F \mapsto E/E_U$ is bijective. It is G-equivariant as P is and F is a G-submodule. We can thus define T as $T = P^{-1}|_F$. □

An important consequence of this theorem is that for indecomposable affine representations of a nilpotent group the affine representation is always unipotent [FGH81].

1.2 Unipotent Representations and Completeness

1.2.1 Expansions in the linear holonomy

The aim of this section is to show that for a compact affine manifold, the holonomy group being unipotent is equivalent to completeness of the affine structure. Of geometrical importance to this will be the concept of an *expansion* in the linear holonomy.

Definition 1.12 (Expansion)**.** Let $T : V \mapsto V$ be a linear map. Then T is said to be an expansion of V if for every eigenvalue λ of T we have $|\lambda| > 1$.

We now wish to prove the following technical lemma which holds in the absence of an expansion.

Lemma 1.13. Let G be a nilpotent group and E a G-module. Suppose that the linear holonomy group does not contain an expansion of E. Then for each $n \in \mathbb{N}_{>0}$, there is a C^r map $\varphi : E \mapsto \mathbb{R}$ which satisfies the following.

1. $\varphi > 0$ almost everywhere.
2. φ is G-invariant.
3. There exists some $a > 0$ such that
$$\varphi(e^{tx}) = e^{ta}\varphi(x)$$
for all $t \in \mathbb{R}$ and $x \in E$.

Before we can do this we will recall some results from algebraic group theory and the representation theory of Lie Algebras

1.2.2 Algebraic groups and representation theory of Lie algebras

Definition 1.14 (Algebraic Group)**.** An *algebraic group* is a matrix group that can be defined using polynomials.

If G is an algebraic group, then there is a unique irreducible component passing through the identity denoted by G°.

Remark 1.15. This unique irreducible component is called the *identity component* of G.

Proposition 1.16. Let G be an algebraic group. Then G° is a normal subgroup of finite index in G whose cosets are connected and irreducible components of G.

Proof. See "Linear Algebraic Groups - Humphreys (7.3)" □

TODO: Flesh these algebraic group proofs out more.

Proposition 1.17. Let G be an algebraic group and H a normal subgroup. Then
$$\overline{[G, H]} = [\overline{G}, \overline{H}]$$
where $\overline{H}$ denotes the Zariski closure.

Proof. Proposition (2.4?) Borel-Linear-Algebraic-Groups https://www.math.utah.edu/~ptrapa/math-library/borel/%20Borel-Linear-Algebraic-Groups-1991.pdf ☐

This proposition leads to the following corollary.

Corollary 1.18. Let G be an algebraic group and N a normal subgroup that is nilpotent. Then $\overline{N}$ is also nilpotent.

Proof. By Proposition 1.17, a central series for N will pass to a central series of $\overline{N}$. ☐

The next result we will state is an important result from representation theory of Lie Algebras. It will guarantee us a basis for our representation such that the matrices of the representation are upper triangular.

Definition 1.19 (Weight Space)**.** Let $\mathfrak{g}$ be a Lie algebra and $\pi : \mathfrak{g} \mapsto \mathfrak{gl}_V$ be a representation on a vector space V. For $\lambda \in \mathfrak{g}^*$ we define the weight space of $\mathfrak{g}$ with respect to λ to be

$$V_\lambda^{\mathfrak{g}} = \{v \in V | \pi(g)v = \lambda(g)v, \quad \forall g \in \mathfrak{g}\}$$

Remark 1.20. If $V_\lambda^{\mathfrak{g}} \neq 0$ we will say that λ is a weight for π.

Lemma 1.21 (Lie's theorem)**.** Let $\mathfrak{g}$ be a solvable Lie algebra and let π be a representation of $\mathfrak{g}$ on a finite dimensional, non-zero vector space V over an algebraically closed field of characteristic zero. Then there exists a weight $\lambda \in \mathfrak{g}^*$ for π.

Proof. See the following https://math.mit.edu/classes/18.745/Notes/Lecture_5_Notes.pdf ☐

Corollary 1.22. Let V be a vector space over an algebraically closed field of characteristic 0. If $\pi : \mathfrak{g} \mapsto \mathfrak{gl}(V)$ is a finite dimensional representation of a solvable Lie algebra, then there is a basis of V such that all linear transformations in $\pi(\mathfrak{g})$ are represented by upper triangular matrices.

We will need a slightly stronger theorem than Lie's theorem which gives a decomposition of our vector space into generalized weight spaces.

Definition 1.23 (Generalized Weight Space)**.** Let π be a representation of a Lie algebra $\mathfrak{g}$ over a finite dimensional vector space V. Then for $\lambda \in \mathfrak{g}^*$, we can associate to λ

$$V_\lambda = \{v \in V | (\pi(g) - \lambda(g))^n v = 0 \text{ for all } g \in \mathfrak{g}\}$$

where n depends on $g \in \mathfrak{g}$ and $v \in V$. If $V_\lambda \neq \{0\}$ then we call it the *generalized weight space* associated to λ.

Remark 1.24. This tells us that $\pi(g) - \lambda(g)$ is a nilpotent operator so that $\pi(g) = \lambda(g) + N(g)$ with $N(g)$ nilpotent.

Lemma 1.25. Let $\mathfrak{g}$ be a nilpotent Lie algebra over $\mathbb{C}$. Let π be a representation of $\mathfrak{g}$ on a finite dimensional complex vector space V. Then there are finitely many generalized weight spaces V_{λ_k}, which are invariant under the action of the representation and $V = \oplus_k V_{\lambda_k}$

Proof. See Proposition 3.2 in https://math.uchicago.edu/~may/REU2013/REUPapers/Ward.pdf □

We now have the required machinary to start our proof.

Lemma 1.26. Let G be a nilpotent group and E a G-module. Suppose that the linear holonomy group does not contain an expansion of E. Then for each $n \in \mathbb{N}_{>0}$, there is a C^r map $\varphi : E \mapsto \mathbb{R}$ which satisfies the following.

1. $\varphi > 0$ almost everywhere.
2. φ is G-invariant.
3. There exists some $a > 0$ such that
$$\varphi(e^{tx}) = e^{ta}\varphi(x)$$
for all $t \in \mathbb{R}$ and $x \in E$.

Proof. If we can prove this for a normal subgroup G_0 of finite index, then we will be able to construct our map φ as follows. Let $g_1G_0, \ldots g_kG_0$ be the left cosets of G_0 and φ_0 a map satisfying properties 1 and 3 defined on G_0 . Now define

$$\varphi(x) = \sum_{i=1}^{k} \varphi_0(g_i x)$$

This will now be G-invariant as G_0 is a normal subgroup multiplication by an element of G will permute the cosets.

As we want to eventually apply apply some results from complex Lie Theory, we complexify the dual space E^* as $F = \mathbb{C} \oplus E^*$. The contragradient representation of G on E^* extends to a representation on F in the obvious way. We will denote this as $\rho : G \mapsto GL(F)$. Now, $\rho(G)$ is a nilpotent subgroup of $GL(F)$ that passes through the identity.

Let H be the identity component of the algebraic closure of $\rho(G)$. As $\rho(G)$ is nilpotent, by Corollary 1.18 so is its Zariski closure and hence its identity component as subgroups of nilpotent groups are nilpotent. H will be a normal subgroup of finite index by Proposition 1.16 and so $G_0 = \rho^{-1}(H)$ will also be a normal subgroup of finite index.

We can now obtain a Lie algebra representation of the Lie algebra of H on F by

$$di : \mathfrak{h} \mapsto \mathfrak{gl}(F)$$

where i is the inclusion map. Importantly, by construction on the underlying vector spaces $\rho(G_0) \subseteq di(\mathfrak{h})$. As H is a nilpotent Lie group, its Lie algebra is nilpotent hence solvable so we can apply Lemma 1.25 to obtain a decomposition $F = \oplus_{k=1}^m F_k$ which is $di(h)$-invariant and hence ρ-invariant by construction when restricted to G_0.

Again, Lemma 1.25 yields that there is a basis B_k of each F_k that represents the operators $di(h)|_{F_k}$ as upper triangular complex matrices

$$di_k(h) = \lambda_k(h)I + N_k(h)$$

where di_k is the restriction to F_k and $N_k(h)$ is strictly upper triangular nilpotent matrix and $\lambda_k(h)$ are the eigenvalues of $\rho(h)$. The first basis vector in each B_k is killed by the strictly upper triangular matrix so that if $f_k \in F_k$ is the first basis vector in B_k

$$di_k(h)f_k = \lambda_k(h)f_k$$

for all $h \in H$.

Now we define the following group morphisms

$$\varphi_k : G_0 \mapsto \mathbb{R}$$
$$\varphi_k(g) = log|\lambda_k(g)|$$

for $k = 1, \ldots, m$.

This defines a map into $\mathbb{R}^m$ by

$$\varphi = (\varphi_1, \ldots, \varphi_m) : G_0 \mapsto \mathbb{R}^m$$

If G_0 does not contain any expansions of E then the image $\varphi(G_0)$ is disjoint from the set

$$P = \{y \in \mathbb{R}^m : y_k > 0, k = 1, \ldots, m\}.$$

As it is a subgroup, it follows that its linear span $span\{\varphi(G_0)\}$ is also disjoint from P as if a point in the image was in the strictly negative orthant, it's inverse would lie in P. Therefore there exists a vector in the orthogonal complement of $\varphi(G_0)$ that lies in the closure of P. We denote this by $v = (v_1, \ldots, v_m)$ and it satisfies

$$v_k \geq 0 \text{ for all } k,$$

$$\sum_{k=1}^{m} v_k = a \geq 0$$

$$\sum_{k=1}^{m} v_k \varphi_k(g) = 0 \text{ for all } g \in G_0$$

If we scale our v by a positive scalar $c > 0$ it will satisfy the same properties so that we can arrange that

$$v_k = 0 \text{ or } v_k > 2r \text{ for each k with } r \text{ coming from the lemma.}$$

Now for each k we have a vector $f_k \in Hom(\mathbb{C} \otimes E, \mathbb{C})$ satisfy $di_k(h)f_k = \lambda_k(h)f_k$. We now include E into $\mathbb{C} \otimes E$ and construct the map

$$\psi : E \mapsto \mathbb{R}$$

$$\psi(x) = \prod_{k=1}^{m} |f_k(x)|^{v_k}$$

Now ψ is C^r differentiable as $|f_k(x)|^{v_k} = (u_k(x)^2 + w_k(x)^2)^m$ with $m > r$. Properties 1) and 3) are satisfied by our construction of v_k. We now show that it is G_0-invariant.

$$\psi(g^{-1}x) = \prod_{k=1}^{m} |f_k(g^{-1}x)|^{v_k}$$

noting this is the contragradient representation and also that f_k is an eigenvector

$$\psi(g^{-1}x) = \prod_{k=1}^{m} |g \circ f_k(x)|^{v_k} = \prod_{k=1}^{m} |\lambda(g) f_k(x)|^{v_k}$$

$$= \prod_{k=1}^{m} |\lambda_k(g)|^{v_k} \prod_{k=1}^{m} |f_k(x)|^{v_k}$$

Now, recalling that $\varphi_k(g) = log|\lambda_k(g)| \Rightarrow e^{\varphi_k(g)} = |\lambda_k(g)|$ so we obtained

$$\prod_{k=1}^{m} |\lambda_k(g)|^{v_k} \prod_{k=1}^{m} |f_k(x)|^{v_k} = e^{\sum_{k=1}^{m} v_k \varphi_k(g)} \psi(x) = \psi(x)$$

completing the proof. $\square$

1.2.3 Consequences of expansions in linear holonomy

The previous theorem leads to several important results regarding completeness of affine manifolds.

Theorem 1.27. Let M be a compact affine manifold. Let $E_0 \subset E$ be a proper linear subspace invariant under the affine holonomy and denote by Λ the linear holonomy. If the image of the linear holonomy group $\Lambda_1 \subseteq GL(E/E_0)$ is nilpotent, then some element of Λ_1 expands E/E_0.

Proof. Let $q : E \mapsto E/E_0$ be the canonical projection. Denote by R the radiant vector field on E/E_0. If $\{y_1, \ldots, y_m\}$ are coordinates on E/E_0, then this appears as $R = \sum_i y_i \frac{\partial}{\partial y_i}$. Then in each affine chart there is a vector field X_i that is π-related to R. As M is compact, we can cover it in finitely many charts and using a partition of unity subordiante to that covering form the vector field

$$X = \sum_{\mu_i} \mu_i X_i$$

on M.

Now suppose that Λ_1 does not contain any expansions of E/E_0. Then by Lemma 1.26, there is a Λ_1-invariant C^1 map $\Psi : E/E_0 \mapsto \mathbb{R}$. As the flow along the radiant vector field at a point z is given by $\phi_t(z) = e^t z$, our map satisfies

$$d\Psi_z R(Z) = \frac{d}{dt} \psi(e^t z)|_{t=0} = \frac{d}{dt} e^{ta} \psi(z)|_{t=0} = a\Psi(z)$$

for some $a > 0$ and all $z \in E/E_0$.

Now the composition

$$\tilde{f} : \tilde{M} \mapsto E \mapsto E/E_0 \mapsto \mathbb{R}$$

is invariant under deck transformations. Indeed, let $\gamma \in \pi_1(M)$ then for $x \in \tilde{M}$

$$\Psi \circ \pi \circ \mathrm{dev}(\gamma \star x) = \Psi \circ \pi \circ \mathrm{hol}(\gamma) \circ \mathrm{dev}(x)$$

and this is invariant as E_0 is invariant under the holonomy group.

This implies that $\tilde{f}$ covers a map $f : M \mapsto \mathbb{R}$. As by Proposition 0.8, we can give charts in terms of the development pair, in affine coordiantates we have that $f = \Psi \circ \pi$. Now

$$\begin{aligned} df_y X(y) = d(\Psi \circ \pi)_y X(y) &= \frac{d}{dt} \Psi \circ \pi(\gamma(t))|_{t=0} \\ &= \frac{d}{dt} \Psi(e^t \tilde{y})|_{t=0} = a\Psi(\tilde{y}) \end{aligned}$$

where $\tilde{y} = \pi(y)$ and $\gamma(t)$ is an integral curve of X starting at y. This gives that

$$df_y X(y) = af(y)$$

It follows that if $\alpha : I \mapsto M$ is an integral curve of X starting at $p \in M$ then

$$\begin{aligned} f(\alpha(t)) = \Psi \circ \pi(\alpha(t)) = \Psi(\overline{\alpha(t)}) &= \Psi(e^t \tilde{p}) = e^{ta}\Psi(\tilde{p}) \\ &= e^{ta} f(\alpha(c)) \end{aligned}$$

Now as $\Psi > 0$ almost everywhere, there exists $x_0 \in M$ with $f(x_0) > 0$ and the integral curve α_0 through x_0 is defined for all t as M is compact. Then $\lim_{t\to\infty} f(\alpha_0(t)) = \lim_{t\to\infty} e^{ta} f(\alpha_0(c)) \to \infty$. But as f is a continuous function and M is compact it must be bounded and so we reach a contradiction. □

This proof leads to several important results. The first result follows almost immediately.

Corollary 1.28. Let M be a compact radiant manifold with nilpotent linear holonomy Λ. Then Λ contains an expansion of E.

Proof. We can take $E_0 = 0$ in Theorem 1.27 and the result follows immediately. □

The next corollary follows from considering the Fitting submodule of E.

Corollary 1.29. Let M be a compact affine manifold with nilpotent affine holonomy group. If $F \neq 0$ then some element of the linear holonomy group expands F.

Proof. Recall from Lemma 1.10, that nilpotence of the affine holonomy gives a unique decomposition $E = E_U \oplus F$ which is invariant under the action of the holonomy group. If we take $E_0 = 0$ in Theorem 1.27, then $F \cong E/E_U$ as a module of the linear holonomy group. □

1.3 Parallel volume

In (TODO:: Markus conjecture paper) it is conjectured that parallel volume is equivalent to completeness for affine manifolds. Using the previous results, we can now give a partial resolution of this conjecture for the case that our manifold has nilpotent holonomy. Importantly, orientable integral affine manifolds always possess a parallel volume form and so the following results will prove important in our classification of such structures.

Theorem 1.30. ([FGH81]) Let M be a compact affine manifold with nilpotent affine holonomy group. Suppose that there is a parallel volume form on M. Then the linear holonomy is unipotent.

Proof. Again we will exploit that there is a unique decomposition $E = E_U \oplus F$ by Lemma 1.10. Now let g be any element of the linear holonomy group. As there is a parallel volume form by Lemma 0.27, $det(g) = 1$. But by the unique decomposition

$$1 = det(g) = det(g|_{E_U})det(g|_F)$$

But $g|_{E_U}$ is a unipotent operator and so has determinant 1. This forces $det(g|_F) = 1$ so that F cannot contain an expansion. But now Corollary 1.29 forces $F = 0$ and so $E = E_U$. □

This will be used in conjunction with the following theorem in order to give equivalent conditions on the completeness of an affine manifold.

Theorem 1.31. Let M be an affine manifold. If M is compact and has unipotent holonomy then M is complete.

Proof. We pick a basepoint x_0 in M. Let $p : \tilde{M} \mapsto M$ be the universal cover of M with $p(\tilde{x_0}) = x_0$. We now choose a developing map $dev : \tilde{M} \mapsto \mathbb{R}^n$ such that $\mathrm{dev}(\tilde{x_0}) = 0$. Let hol be the holonomy representation and $\mathrm{hol}(\pi_1(M)) = \Gamma$ be the affine holonomy group.

Now, suppose that the linear affine holonomy group is unipotent. Then by **TODO: Humphreys Linear algebraic groups** there exists a linear flag

$$0 = F_0 \subset F_1 \subset \cdots \subset F_n = \mathbb{R}^n$$

preserved by the linear affine holonomy group. Furthermore, the induced action on the quotients F_k/F_{k-1} is trivial and so the restriction to F_k preserves a linear functional $l_k : F_k \mapsto \mathbb{R}$ with kernel F_{k-1}.

This determines a family of parallel fields $\mathfrak{F}_k \subset TM$. In affine coordinates, each of these is spanned by a subset of the coordinate vector fields and so it is clear that they are integrable.Furthermore, each linear map l_i determines a parallel 1-form ω_i defined on $\mathfrak{F}_i$ and vanishing on $\mathfrak{F}_{i-1}$.

We can construct vector fields X_i on M, tangent to $\mathfrak{F}_i$ such that $X_i(l_i) = 1$ using partitions of unity. These lift to vector fields $\tilde{X}_i$ on the universal cover.

We first show surjectivity of the developing map. Let $\gamma_i(t)$ be an integral curve of $\tilde{X}_i$ starting at the point x_i. As dev is a local diffeomorphism we have that

$$\frac{d}{dt}\mathrm{dev}\circ\gamma_i(t) = dev_*\circ\tilde{X}(\gamma_i(t))$$
$$\Rightarrow \mathrm{dev}\circ\gamma_i(t) = dev_*\circ\tilde{X}_i(\gamma_i(t))t + \mathrm{dev}\circ\gamma_i(0)$$

applying our linear functional l_i to this we obtained

$$\begin{aligned} l_i(\mathrm{dev}\circ\gamma_i(t)) &= l_i\circ dev_*(\tilde{X}_i(\gamma_i(t)))t + l_i(\mathrm{dev}\circ\gamma_i(0)) \\ &= l_i(\tilde{X}_i)t + l_i(\mathrm{dev}\circ\gamma_i(0)) = t + l_i(\mathrm{dev}\circ\gamma_i(0)) \end{aligned} \tag{*}$$

Now we can start our inductive step. Taking $i = n$ we start at $\tilde{x}_0 \in \tilde{M}$ such that $\mathrm{dev}(\tilde{x}_0) = 0$. We flow along $\tilde{X}_n$ for time $t = l_n(v)$ until the point $\tilde{x}_1 \in \tilde{M}$ with $\mathrm{dev}(\tilde{x}_1) = v_1$. From (*) we obtain

$$l_n(v_1) = l_n(v) + 0$$
$$\Rightarrow v - v_1 \in \ker l_n = F_{n-1}$$

Now, we continue e.g. flow along $\tilde{X}_{n-1}$ for $t = l_{n-1}(v - v_1)$ ending at $\tilde{x}_2$ with $\mathrm{dev}(\tilde{x}_2) = v_2$. Then

$$l_{n-1}(v_2) = l_{n-1}(v - v_1) + l_n(v_1)$$
$$\Rightarrow l_{n-1}(v - v_2) = 0 \Rightarrow v - v_2 \in \ker l_{n-1} = F_{n-2}$$

We continue this until we reach $F_0 = \{0\}$ showing surjectivity.

Now for injectivity we consider a path

$$\gamma_0 : [a, b] \mapsto \tilde{M}$$

which develops to a closed loop $\delta : [a, b] \mapsto \mathbb{R}^n$. We can deform γ_0 to a new path γ_1 such that it develops entirely inside F_{n-1}. We now define a new path $\gamma_t : [a, b] \mapsto \tilde{M}$ where $\gamma_t(s)$ is the image of $\gamma_0(s)$ at time $-tl_n(\delta(s))$ under the flow of $\tilde{X}_n$. Applying dev gives that

$$\mathrm{dev}\circ\gamma_t(s) = \mathrm{dev}(\gamma_0(s)) - tl_n(\delta(s))dev_*(\tilde{X}(\gamma_0(s)))$$
$$\Rightarrow l_n(\mathrm{dev}\circ\gamma_t(s)) = l_n(\mathrm{dev}(\gamma_0(s))) - tl_n(\delta(s))l_n\circ dev_*(\tilde{X}(\gamma_0(s)))$$
$$\Rightarrow l_n(\mathrm{dev}\circ\gamma_t(s)) = l_n(\mathrm{dev}(\gamma_0(s))) - tl_n(\delta(s))$$

At $t = 1$ this yields that $\mathrm{dev}\circ\gamma_t(s) \in \ker l_n = F_{n-1}$. Repeating this process we eventually get a curve that develops entirely in $F_0 = \{0\}$ showing injectivity. □

We summarize the previous results into the following theorem.

Theorem 1.32. Let M be a compact affine manifold with nilpotent affine holonomy representation. Then the following are equivalent

1. M is a complete affine manifold.
2. The linear holonomy is unipotent.
3. M has a parallel volume form.

Proof. We have that 2) ⇒ 1) by Theorem 1.31. We have that 3) ⇔ 2) by Theorem 1.30. □

Remark 1.33. We note that if the fundamental group of M is nilpotent, then its holonomy representation will also be nilpotent as this is preserved by group morphisms.

CHAPTER TWO

Lagrangian Fibrations and Integral Affine Manifolds

2.1 Symplectic Geometry

In this section we will recall some of the basics of symplectic geometry and Lagrangian submanifolds.

Definition 2.1 (Symplectic Manifold)**.** A symplectic manifold (M, ω) consists of a smooth manifold M and a closed 2-form ω such that ω is non-degenerate.

Symplectic geometry has its roots in classical mechanics as the phase space of a dynamical system naturally has the structure of a symplectic manifold. TODO: elaborate more on symplectic geometry - Hamiltons equations, applications to topology, Poisson geometry.

Non-degeneracy of the symplectic form gives a canonical way to identify the tangent and cotangent bundle of a symplectic manifold. This is done via the map

$$\tilde{\omega}|_p : T_pM \to T_p^*M$$
$$v \mapsto \omega_p(v, \cdot)$$

for $v \in T_pM$. This is an isomorphism by non-degeneracy and so, namely, it is invertible. This identification leads to the notion of a Hamiltonian vector field.

Definition 2.2 (Hamiltonian Vector Field)**.** For any $H \in C^\infty(M)$ we can associate a unique Hamiltonian vector field by

$$\iota_{X_H}\omega := \omega(X_H, \cdot) = dH$$

For mathematicians there is generally no distinguished choice of Hamiltonian function H. However, physicists often pick a Hamiltonian function based on physical criteria and are generally interested in studying the flow of this Hamiltonian vector field which, for instance, recovers Hamilton's equations.

2.1.1 The Cotangent Bundle

One of the most important examples of symplectic manifolds is the cotangent bundle, which naturally inherits the structure of a symplectic manifold. In fact, Darbouxs theorem states that every symplectic manifold looks locally like a cotangent bundle. This class of symplectic manifolds will playu a big role throughout this thesis and we will see now exactly how the cotangent bundle obtains a symplectic structure.

Example 2.3 (Cotangent Bundle)**.** Let Q be any smooth manifold and $M = T^*Q$ its cotangent bundle. We will now illustrate that there is a canonical one form $\theta \in \Omega^1(T^*Q)$ and show that the exterior derivative $-d\theta$ defines a symplectic structure on T^*Q.

Let $\pi : T^*Q \to Q$ be the canonical projection map. Then for $X_p \in T_p(T^*Q)$ we define $\langle \theta_p, X_p \rangle := \langle p, d_p\pi(X_p) \rangle$ where $\langle \cdot, \cdot \rangle$ denotes contraction of a vector and covector. This canonical one-form satisfies the following important property.

Lemma 2.4. θ is the unique one-form such that for any other one-form $\alpha \in \Omega^1(Q)$

$$\alpha = \alpha^*\theta$$

where we view α as a section $\alpha : Q \to T^*Q$.

Proof. Let $w \in T_qQ$. Then we have that

$$\begin{aligned} \langle (\alpha^*\theta)_q, w \rangle &= \langle \theta_{\alpha_q}, d_q\alpha(w) \rangle \\ &= \langle \alpha_q, w \rangle \end{aligned}$$

Uniqueness is obtained by noting that every tangent vector $v \in T_p(T^*Q)$ can be written in the form $d_q\alpha(w)$ except for those in the kernel of $d_p\pi$. But the vectors not in this kernel form an open, dense set and so by continuity we get uniqueness. □

We can also find an expression for this canonical one-form in coordinates.

Lemma 2.5. In local cotangent coordinates $q_1, \dots, q_n, p_1, \dots, p_n$ on T^*Q, the canonical one-form is given by

$$\theta|_{T^*U} = \sum_j p_j dq_j$$

Proof. We let $\alpha = \sum_j \alpha_j dq_j$ be a 1-form on U. We have that $\alpha^* p_j = \alpha_j$ and $\alpha^* q_j = q_j$ which gives that

$$\begin{aligned} \alpha^*\theta|_{T^*U} &= \alpha^* \sum_j p_j dq_j \\ &= \sum_j \alpha_j dq_j \\ &= \alpha \end{aligned}$$

□

Putting this together we obtain a symplectic form on T^*Q.

Theorem 2.6. The 2-form $-d\theta$ defines a symplectic form on T^*Q.

Proof. In local coordinates we have that $\omega = \sum_j dq_j \wedge dp_j$ which is clearly closed and non-degenerate. □

We will also wish to speak about maps between the symplectic manifolds. The correct notion of "isomorphism" in this category is the following.

Definition 2.7. Let (M, ω) and (N, α) be two symplectic manifolds. Then a symplectomorphism between M and N is a diffeomorphism

$$\varphi : M \to N$$

such that $\omega = \varphi^* \alpha$.

2.1.2 Lagrangian Submanifolds

The goal of this thesis is to classify Lagrangian fibrations over compact surfaces. In this section we will introduce the concept of Lagrangian submanifolds and prove some basic results regarding these.

Definition 2.8 (Symplectic Orthogonal)**.** Let (M, ω) be a symplectic manifold and let $N \subset M$ be a submanifold. The symplectic orthogonal at a point $p \in M$ is

$$(T_pN)^{\omega} := \{v \in T_pM \mid \omega(v, w) = 0, \forall w \in N\}$$

- If $(T_pN)^{\omega} \subset T_pN$ then T_pN is said to be *co-isotropic*.
- If $T_pN \subset (T_pN)^{\omega}$ then T_pN is said to be *isotropic*.
- If $(T_pN)^{\omega} = T_pN$ then T_pN is said to be *Lagrangian*.
- If $(T_pN)^{\omega} \cap T_pN =$ then T_pN is said to be *symplectic*.

A submanifold N of a symplectic manifold M is said to be co-isotropic (isotropic, Lagrangian, symplectic) if its tangent space is a co-isotropic (isotropic, Lagrangian, symplectic) subspace of T_pM at every point $p \in M$.

Remark 2.9. We note that a submanifold $\iota : N \to M$ is isotropic if $\iota^*\omega = 0$ and furthermore, it is Lagrangian if we also have $\dim N = \frac{1}{2}\dim N$.

We will make use of the following lemma.

Lemma 2.10. The graph $\Gamma_\alpha \subset T^*Q$ of a 1-form α is Lagrangian if and only if α is closed.

Proof. It is clear that $\dim \Gamma_\alpha = \frac{1}{2}\dim T^*Q$.

$$\begin{aligned}\alpha^*\omega &= -\alpha^* d\theta \\ &= -d\alpha^*\theta \\ &= -d\alpha\end{aligned}$$

and so $\alpha^*\omega = 0$ if and only if α is closed completing the proof by the previous remark. $\square$

The cotangent bundle T^*Q also gives rise to another Lagrangian submanifold.

Example 2.11. Let $\pi : T^*Q \to Q$ and fix $x \in Q$. Then $\pi^{-1}(x) = T_x^*Q$ is a Lagrangian submanifold inside (T^*Q, ω). It is easy to see that $\dim T_x^*Q = \frac{1}{2}\dim T^*Q$. Letting $i : T_x^*Q \hookrightarrow T^*Q$ be the inclusion and considering local coordinates $(q_1, \ldots, q_n, p_1 \ldots p_n)$ around x and writing $\omega = \sum_j dq_j \wedge dp_j$ it follows that $\omega|_{T_x^*Q} = 0$ as $i^*(\sum_j dq_j \wedge dp_j) = \sum_j d(i^*q_j) \wedge d(i^*p_j)$ and $d(i^*q_i) = 0$ as i^*q_i is constant.

2.2 Lagrangian Fibrations

One of the goals of this thesis is to classify the Lagrangian fibrations over closed surfaces. In this section we will give an overview of these objects and give some examples and basic results surrounding them. We will then describe how the base space of a Lagrangian fibration inherits an integral affine structure.

2.2.1 Definition of Lagrangian Fibration

Definition 2.12. A Lagrangian fibration $\pi : (M, \omega) \to B$ is a surjective map whose regular fibres are Lagrangian submanifolds of (M, ω).

Example 2.13. We seen in Lemma 2.10 that the fibres of $\pi : (T^*Q, \omega_{can}) \to Q$ are Lagrangian submanifolds and hence this is a Lagrangian fibration.

Example 2.14. Let $Q = (\mathbb{R}/\mathbb{Z})^n = \mathbb{T}^n$ be the n-torus. Then we have that $T^*(\mathbb{T}^n) = \mathbb{T}^n \times \mathbb{R}^n$ and the map

$$\pi : T^*(\mathbb{T}^n) \to \mathbb{R}^n$$

is a Lagrangian fibration with fibre $\mathbb{T}^n$.

Definition 2.15 (Shift by a 1-form.)**.** Let $\pi : (T^*Q, \omega_{can}) \to Q$ and fix $\alpha \in \Omega^1(Q)$. We define the *shift by* α to be the map

$$s_\alpha : T^*Q \to T^*Q$$

$$\beta_q \mapsto \beta_q + \alpha_q$$

Proposition 2.16. This defines a symplecto-morphism of T^*Q if and only if $d\alpha = 0$.

Proof. This map is a fibrewise translation and hence is obviously a diffeomorphism. We now check when it preserves the symplectic form. We calculate

$$\begin{aligned} s_\alpha^*\omega_{can} &= -s_\alpha^* d\theta \\ &= -ds_\alpha^*\theta \end{aligned}$$

We calculate

$$\begin{aligned} s_\alpha^*\theta(v)|_{\beta_q} &= \theta_{\beta_q+\alpha_q}(ds_\alpha(v)) \\ &= (\beta_q + \alpha_q)(d(\pi \circ s_\alpha)(v)) \\ &= (\beta_q + \alpha_q)(v) \\ &= \theta_{\beta_q}(v) + \alpha_q(v) \end{aligned}$$

and so $s_\alpha^*\theta = \theta + \alpha$. Now

$$\begin{aligned} -ds_\alpha^*\theta &= -d\theta - d\alpha \\ &= \omega_{can} - d\alpha \end{aligned}$$

and hence this defines a symplecto-morphism if and only if $d\alpha = 0$. $\square$

2.2.2 Integral Affine Structure on the Base

2.2.3 Action Angle Coordinates

CHAPTER THREE

Classification of Integral Affine Closed Surfaces

3.1 Some basic results

We first discuss our approach to the classification problem.

We make use of the following theorem from Milnor relating to the existence of a flat connection on closed orientable surfaces. We can handloe the non-orientable case by applying the following classification theorem.

Theorem 3.1. Every compact, connected 2-manifold is homeomorphic to the sphere, the connected sum of tori or the connected sum of $\mathbb{R}P^2$.

Proof. Hatcher? □

Theorem 3.2. Let M^2 be a closed orientable surface of genus g. Then if $g \geq 2$ it does not possess a connection with zero curvature.

Proof. cite "On the existence of a connection with zero curvature" by Milnor. □

Corollary 3.3. The possible orientable closed surfaces admitting an integral affine structure are the sphere S^2 or the torus $\mathbb{T}^2$.

Proof. This follows from the classification of closed 2 surfaces. (TODO: get source - probably hatcher or something) □

However, the following simple argument shows that the sphere S^2 does not admit such a structure.

Proposition 3.4. The sphere S^2 does not admit an affine structure.

Proof. Suppose there was such an integral affine structure. Then

$$\mathrm{dev} : S^2 \to \mathbb{R}^2$$

is a local diffeomorphism and hence an open map. However dev is also continuous, and so $\mathrm{dev}(S^2)$ is a compact subset of $\mathbb{R}^2$ and hence closed. Now, as $\mathrm{dev}(S^2) \neq \emptyset$, by connectivity $\mathrm{dev}(S^2) = \mathbb{R}^2$ but $\mathbb{R}^2$ is not compact. ☐

We have the following immediate corollary.

Corollary 3.5. The only closed, orientable surface admitting an (integral) affine structure is $\mathbb{T}^2$.

We will deal with the non-orientable case by passing to the orientable double cover.

Lemma 3.6. Every closed, non-orientable surface has a double cover that is orientable. Furthermore, the genus of the double cover is given by

$$g = \bar{g} - 1$$

where $\bar{g}$ is the number of connected sums of $\mathbb{R}P^2$.

This lemma leads to the following corollary.

Corollary 3.7. The only closed surfaces admitting an (integral) affine structure are the torus $\mathbb{T}^2$ and the Klein bottle $\mathbb{K}^2$.

Proof. As any covering space of an integral affine structure inherits an integral affine structure by (TODO: do this in (G,X)-structure section), it follows that the orientable double cover of a non-orientable integral affine surface must be $\mathbb{T}^2$. Furthermore, it must have non-orientable genus (TODO: define this somewhere?) $\bar{g} = 1$ or 2. If $\bar{g} = 1$ our surface is homeomorphic to $\mathbb{R}P^2$ which is covered by S^2 and hence admits no such structure. If $\bar{g} = 2$ we get the Klein bottle which is covered by $\mathbb{T}^2$ and hence can admit such a structure. ☐

3.2 Classification of Integral Affine structures on the torus

3.2.1 Completeness and freeness of the action

We now look at how we can classify the integral affine structures on the torus. The following observation will greatly simplify the classification process.

Lemma 3.8. Any integral affine structure on the torus is complete.

Proof. We have that the fundamental group $\pi_1(\mathbb{T}^2) = \mathbb{Z} \times \mathbb{Z}$ is abelian and hence nilpotent. As the torus is orientable, our linear holonomy group $\Gamma \subseteq \mathrm{SL}(2, \mathbb{Z})$ by (TODO: Reference Corr 0.26). But now by (TODO: finish nilpotent section e.g. combine to theorem). ☐

(TODO: This might be a good place for discussion on completeness stuff. Auslander-Markus show that every complete affine manifold is quotient by linear holonomy).

From the above discussion, classifying complete integral affine structures on a surface M amounts to classifying injective homomorphisms from the fundamental group into the group $\mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ which induce a free and proper action on $\mathbb{R}^2$.

Lemma 3.9. If $g = (A, b) \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ acts freely on $\mathbb{R}^2$, then

$$\text{if } \det(A) = 1 \text{ then } \mathrm{Tr}(A) = 2$$

and,

$$\text{if } \det(A) = -1 \text{ then } \mathrm{Tr}(A) = -1$$

Proof. Since the action by g is free, this implies

$$Ax + b = x$$

has no solution. But now,

$$(A - I)x = -b$$

has no solutions for all $x \in \mathbb{R}^2$ and so $\det(A - I) = 0$. Calculating this determinant yields

$$1 - \mathrm{Tr}(A) + \det(A) = 0$$

and substituting in the appropriate values of $\det(A)$ gives the required results.

□

(TODO: I think this result is surprisingly useful for example I think proofs of Auslander conjecture in low dimensions rely on something like this. Perhaps discuss more.)

We note also that 1 is an eigenvalue of such an A as by the above proof $\det(A - 1 \cdot I) = 0$. Using this result we can find a nice basis to express the linear part of our element $g \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$.

Lemma 3.10. Let $g = (A, b) \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ act freely. Then there is an integral basis of $\mathbb{R}^2$ such that

$$A = \begin{pmatrix} 1 & n \\ 0 & \pm 1 \end{pmatrix}$$

Proof. By our previous lemma, A must have 1 as an eigenvalue. We can take an eigenvector e_1 such that e_1 is integral, e.g. its components are integers. Now as $\mathrm{Tr}(A) = 2$ or 0, it follows that we can pick our second basis vector such that

$$A = \begin{pmatrix} 1 & n \\ 0 & \pm 1 \end{pmatrix}$$

since we need $A \in \mathrm{GL}(2, \mathbb{Z})$. Furthermore, by changing orientation if necessary we may assume $n \geq 0$. □

Lemma 3.11. Let $H \subset \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)^+$ be a subset of the elements of determinant one and suppose H acts freely on $\mathbb{R}^2$. Then there is an integral basis such that for all $(A, b) \in H$ we can write

$$A = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$$

for $n \in \mathbb{Z}$

Proof. Let $(A, a), (B, b) \in H$. Now by Lemma 3.10, we can write

$$A = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

with $n \neq 0$. Now by Lemma 3.9, we have that

$$\begin{aligned} \mathrm{Tr}(A^k B) &= a + nck + d \\ &= \mathrm{Tr}(B) + nck \\ &= 2 \end{aligned}$$

As $n \neq 0$ and this holds for all $k \in \mathbb{Z}$, it follows that $c = 0$.

But now, $\det B = ad = 1$ and $\mathrm{Tr}(B) = a + d = 2$ and so $a = d = 1$ giving the required result as $b \in \mathbb{Z}$. ☐

We have the following corollary of this result.

Corollary 3.12. For any two elements $(A, a), (B, b) \in H$, there exists relatively prime $(k, l) \neq (0, 0)$ such that $A^k B^l = 1$.

Proof. We simply take $k = \frac{m}{(m,n)}$ and $l = \frac{-n}{(m,n)}$ and then

$$\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}^k \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix}^l = 1$$

. ☐

3.2.2 Finding the possible generators

We have now built up the required machinery to begin the classification of the integral affine structures on the torus $\mathbb{T}^2$.

Theorem 3.13. Any complete integral affine structure on $\mathbb{T}^2$ has its affine holonomy group generated by one of the following subgroups of $\mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$

1. $\mathcal{A}_{a,b,c,d}$ - These integral affine structures are the quotient of $\mathbb{R}^2$ by a lattice of pure translations. The generators in $\mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ are given by

$$\left(1, \begin{pmatrix} a \\ c \end{pmatrix}\right) \text{ and } \left(1, \begin{pmatrix} b \\ d \end{pmatrix}\right)$$

with $a, b, c, d \in \mathbb{R}$ and $ad - bc \neq 0$.

2. $\mathcal{B}_{n,a,b}$ - The generators in $\mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ are given by

$$\left(1, \begin{pmatrix} a \\ 0 \end{pmatrix}\right) \text{ and } \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ b \end{pmatrix}\right)$$

with $a, b > 0$ and $n \in \mathbb{N}$.

Proof. Let $\pi_1(\mathbb{T}^2) = \langle \tilde{C}, \tilde{D} \rangle$ and denote by C, D the images of these generators under the affine holonomy representation. We wish to show that these generators reduce to the form of either $\mathcal{A}_{a,b,c,d}$ or $\mathcal{B}_{n,a,b}$. We can do this by

1. Changing the generators $\tilde{C}, \tilde{D}$ of the $\pi_1(\mathbb{T}^2)$.
2. Conjugating the images of these generators by some $g \in \mathrm{Aff}^+_{\mathbb{Z}}(\mathbb{R})$ - this corresponds to pre-composing the developing map by g which preserves the affine structure.

If both $C = \left(1, \begin{pmatrix} a \\ c \end{pmatrix}\right)$ and $D = \left(1, \begin{pmatrix} b \\ d \end{pmatrix}\right)$ are pure translations, e.g. the linear component is the identity, injectivity forces that our translational vectors are linearly independent, otherwise $\mathrm{hol}(k\tilde{C} + m\tilde{D}) = 0$ for some $k, m \neq 0 \in \mathbb{R}$. In this case our lattice in $\mathbb{R}^2$ would be one-dimensional. This gives the condition that $ad - bc \neq 0$ and completes the first class of lattices.

Now suppose that C, D are not pure translations. Then by Corollary 3.12, there exists relatively prime $k, l \in \mathbb{Z}$ such that $C^k D^l$ is a translation. We can therefore change generators in $\mathbb{Z}^2$ to $\tilde{A} = \tilde{C}^k \tilde{D}^l$ and $\tilde{B} = \begin{pmatrix} a \\ b \end{pmatrix}$. In the basis $\tilde{C}, \tilde{D}$ we have that the matrix

$$M = \begin{pmatrix} k & a \\ l & b \end{pmatrix}$$

has $\det M = kb - la$ and since k, l are relatively prime by Bezout's theorem we can always pick $a, b \in \mathbb{Z}$ such that this determinant is 1 and hence an automorphism of our lattice.

We denote the images of our new generators under the affine holonomy representation by

$$A' = \left(I, a = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}\right) \text{ and } B' = \left(B, b = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}\right).$$

Now by Lemma 3.11, we can find a basis of $\mathbb{R}^2$ such that $B = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ with $n > 0$. Now $\mathrm{hol}(\pi_1(\mathbb{T}^2))$ is abelian so we require that $A'B' = B'A'$ which gives the condition

$$Ba = a$$

this implies that

$$\begin{aligned} \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} &= \begin{pmatrix} a_1 + na_2 \\ a_2 \end{pmatrix} \\ &= \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \end{aligned}$$

which gives $a_2 = 0$ as $n > 0$ and so $a = \begin{pmatrix} a_1 \\ 0 \end{pmatrix}$. We note that $a_1 \neq 0$ as otherwise the action is not free. For freeness, we also require that $A^l B^k$ acts freely for all $k, l \in \mathbb{Z}$. We check

$$\begin{aligned} \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}^k \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} kb_1 + la_1 \\ kb_2 \end{pmatrix} &= \begin{pmatrix} x + kny + kb_1 + la_1 \\ y + kb_2 \end{pmatrix} \\ &= \begin{pmatrix} x \\ y \end{pmatrix} \end{aligned}$$

If $b_2 = 0$ then there is a fixed point for $y = \dfrac{-kb_1 - la_1}{kn} \in \mathbb{R}$ so we must have $b_2 \neq 0$. Conversely, if $b_2 \neq 0$ and the action is not free, this requires $k = 0$ and so $x + la_1 = a_1$ for

$l \neq 0$ but $a_1 \neq 0$ and hence there are no solutions. Now, to show that this takes the form of $\mathcal{B}_{n,a,b}$, we conjugate by

$$T = \left(I, \begin{pmatrix} 0 \\ \frac{-b_1}{n} \end{pmatrix} \right)$$

this fixes A' and so we compute

$$T^{-1}B'T = \left(B, \begin{pmatrix} 0 \\ b_2 \end{pmatrix} \right)$$

and so we may assume $b_1 = 0$. We can also make the change $A' \mapsto A'^{-1}$ and $B' \mapsto B'^{-1}$ and assume $a_1, b_2 > 0$ which completes the proof. □

3.2.3 When are two structures equivalent?

This gives the possible integral affine structures on the 2-torus. However we still have to tackle the question of when 2 lattices give the same integral affine structure. This occurs when the subgroups of the integral affine group determined by the holonomy representation are conjugate to eachother.

We have the following classification result.

Lemma 3.14. Lattices from different series do not induce equivalent integral affine structures.

Proof. Lattices in series $\mathcal{A}_{a,b,c,d}$ are generated by pure translations and hence cannot be conjugated into lattices of the form $\mathcal{B}_{n,a,b}$.

□

Theorem 3.15. Lattices $\mathcal{A}_{a,b,c,d}$ and $\mathcal{A}_{a',b',c',d'}$ induce equivalent integral affine structures if and only if

$$X = GX'H$$

where $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $X' = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$ with $G, H \in GL(2, \mathbb{Z})$

Proof. Performing lattice automorphisms in $\pi_1(\mathbb{T}^2)$ and conjugating in the image of the generators in $\mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ preserves the integral affine structure.

Let $\pi_1(\mathbb{T}^2) = \langle v_1, v_2 \rangle$ with $\mathrm{hol}(v_1) = \left(I, \begin{pmatrix} a \\ c \end{pmatrix} \right)$ and $\mathrm{hol}(v_2) = \left(I, \begin{pmatrix} b \\ d \end{pmatrix} \right)$ and set $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Let $H \in GL(2, \mathbb{Z})$ and $g = (G, w) \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$. We note that conjugating a pure translation (I, v) by g gives the element (I, Gv). We compute

$$\mathrm{hol}(Hv_1) = \left(I, XH \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right)$$

and

$$\mathrm{hol}(Hv_2) = \left(I, XH \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$$

This corresponds to an automorphism of our lattice in $\pi_1(\mathbb{T}^2)$. We can also conjugate generators by $g = (G, w) \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ which gives

$$g \circ \mathrm{hol}(Hv_1) \circ g^{-1} = \left(I, GXH \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right)$$

and

$$g \circ \mathrm{hol}(Hv_2) \circ g^{-1} = \left(I, GXH \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right)$$

and so it follows that the integral affine structures are equivalent if and only if $X' = GXH$. TODO: finish this proof. ☐

Theorem 3.16. Lattices in $\mathcal{B}_{n,a,b}$ never induce the same integral affine structure.

Proof. Suppose that $\mathcal{B}_{n,a,b} = \langle (1, \bar{a}), (B, \bar{b}) \rangle$ and $\mathcal{B}_{n',a',b'} = \langle (1, \bar{a}'), (B', \bar{b}') \rangle$ induce equivalent integral affine structures. We must have that

$$g\tilde{A}g^{-1} = \tilde{A}'^k \tilde{B}'^r \tag{3.1}$$

$$g\tilde{B}g^{-1} = \tilde{A}'^l \tilde{B}'^s \tag{3.2}$$

where $\tilde{A}, \tilde{B}$ and $\tilde{A}', \tilde{B}'$ are the images of the generators of $\pi_1(\mathbb{T}^2)$ of the structures $\mathcal{B}_{n,a,b}$ and $\mathcal{B}_{n',a',b'}$ and $g = (G = [g_{ij}], \bar{h}) \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$. The matrix

$$M = \begin{pmatrix} k & l \\ r & s \end{pmatrix} \in GL(2, \mathbb{Z})$$

corresponds to an automorphism of our lattice. This implies that

$$GBG^{-1} = \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}$$

where $B = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$. Computing this condition yields

$$\begin{pmatrix} * & * \\ g_{21}g_{22} - ng_{21}^2 - g_{21}g_{22} & * \end{pmatrix} = \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}$$

and so $g_{21}g_{22} - ng_{21}^2 - g_{21}g_{22}$ which gives that $g_{21} = 0$.

Now, inspecting the linear part of $g\tilde{A}g^{-1}$ as the linear part of $\tilde{A}$ is the identity this must also be a translation e.g. $\tilde{A}'^k \tilde{B}'^r$ is a translation. Again, inspecting the linear components we must have that $r = 0$. Our matrix M corresponding to the automorphism of our lattice now looks like

$$M = \begin{pmatrix} k & l \\ 0 & s \end{pmatrix}$$

and so $k, s = \pm 1$ as this matrix needs to have determinant ± 1. Now from Equation 3.1 we obtain

$$\begin{aligned} (G, \bar{h}) \left(I, \begin{pmatrix} a \\ 0 \end{pmatrix}\right) (G^{-1}, -G^{-1}\bar{h}) &= \left(I, G \begin{pmatrix} a \\ 0 \end{pmatrix}\right) \\ &= \left(I, \begin{pmatrix} g_{11}a \\ 0 \end{pmatrix}\right) \\ &= \left(I, \begin{pmatrix} ka' \\ 0 \end{pmatrix}\right) \end{aligned}$$

where the last equality comes from the RHS of Equation 3.1. This now gives that

$$g_{11}a = ka' \Rightarrow g_{11}a = \pm a'$$

as $k = \pm 1$. But $a, a' > 0$ by the classification and so $g_{11} = k$ and $a = a'$.

Next we inspect Equation 3.2. The matrix part gives that

$$GBG^{-1} = B'^{s}$$

Computing this gives

$$\begin{aligned}\begin{pmatrix} g_{11} & g_{12} \\ 0 & g_{22} \end{pmatrix}\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}\begin{pmatrix} g_{11} & -g_{12}g_{11}g_{22} \\ 0 & g_{22} \end{pmatrix} &= \begin{pmatrix} 1 & g_{11}g_{22}n \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & n's \\ 0 & 1 \end{pmatrix}\end{aligned}$$

Where the last equality comes again from the RHS of Equation 3.2. Comparing matrix entries yields $g_{11}g_{22}n = n's$. Using that $g_{11} = \pm 1$, $g_{22} = \pm 1$ and $s = \pm 1$ we have that $n = \pm n'$. But both $n, n' \in \mathbb{N}$ by the classification and so we must have $n = n'$. This now gives that $s = g_{11}g_{22}$. This was the linear part of Equation 3.2 so we now analyze the translational component.

Comparing the translational parts yields

$$-GBG^{-1}\bar{h} + G\bar{b} + \bar{h} = s\bar{b}' + l\bar{a}' \tag{3.3}$$

We can factorise te LHS as

$$-GBG^{-1}h + G\bar{b} + \bar{h} = G(I - B)G^{-1}\bar{h} + G\bar{b}$$

Now we simplify

$$G(I - B)G^{-1}\bar{h} = \begin{pmatrix} x \\ 0 \end{pmatrix}$$

where $x \in \mathbb{R}$ is arbitrary depending on $\bar{h}$. Now Equation 3.3 becomes

$$\begin{pmatrix} x \\ 0 \end{pmatrix} + G\bar{b} = s\bar{b}' + l\bar{a}'$$

Expanding this we are left with

$$\begin{pmatrix} x + g_{12}b \\ g_{22}b \end{pmatrix} = \begin{pmatrix} la' \\ sb' \end{pmatrix}$$

This gives $g_{22}b = sb'$ and again as $b, b' > 0$ and $g_{22}, s = \pm 1$ we have $b = b'$ and now we are finished as we showed $a = a'$, $b = b'$ and $n = n'$.

□

This completes the classification of the integral affine structures on the torus $\mathbb{T}^2$. We note that lattices in the series $\mathcal{A}_{a,b,c,d}$ induce a Euclidean structure on the torus - e.g. their holonomy representation is a subgroup of $O(n) \rtimes \mathbb{R}^n$. This implies that there is a Riemannian metric preserved by this structure. However, the integral affine structures of $\mathcal{B}_{n,a,b}$ never induce a Euclidean structure on the torus.

Next, we will look at the classification of integral affine structures over the Klein bottle, hence completing the classification of all integral affine structures over closed surfaces.

3.3 Classification of Integral Affine structures on the Klein Bottle

3.3.1 Completeness of integral affine structures on K^2

We now wish to classify the integral affine structures on the Klein bottle. We would hope to proceed as before by calculating all of the injective homomorphisms form $\Gamma = \pi_1(K^2) \to \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ that induce a free and proper action on $\mathbb{R}^2$. However, in order to do this we would need to know that every integral affine structure on K^2 is complete. The following lemma takes care of this fact.

Lemma 3.17. Let N be an integral affine manifold and suppose that there is a regular covering $M \to N$ by a compact orientable integral affine manifold whose fundamental group is nilpotent. Then any integral affine structure on N is complete.

Proof. We can pull back the integral affine structure on N to M. A regular covering induces an injection of fundamental groups $\iota : \pi_1(M) \mapsto \pi_1(N)$. We can induce the holonomy representation on M by $\mathrm{hol} \circ \iota$ where hol is the affine holonomy representation induced by the integral affine structure on N. As M and N share a common universal cover X, the integral affine structure on N induces a developing map $\mathrm{dev} : X \mapsto \mathbb{R}^n$ and this induced holonomy representation will remain equivariant with respect to this developing map as the action of $\pi_1(\mathbb{T}^2)$ on the universal cover is a restriction of the action of $\pi_1(K^2)$.

From Proposition 0.8 this defines an integral affine structure on M which has the same developing map as that of N. Therefore, if the integral affine structure on N is incomplete then the integral affine structure on M will also be. However, M is a compact orientable integral affine manifold with nilpotent fundamental group and hence by (TODO: reference theorem) is complete. It follows that any integral affine structure on N must also be complete. $\square$

Throughout this section we let $\pi_1(K^2) = \Gamma$ have presentation

$$\Gamma = \langle a, b \mid aba = b \rangle$$

and we note that $\langle a, b^2 \rangle \subset \Gamma$ is abelian. Indeed, the oriented double cover of K^2 is the torus $\mathbb{T}^2$ and so there is an injection of fundamental groups $\pi_1(\mathbb{T}^2) \hookrightarrow \pi_1(K^2)$. This abelian subgroup can be realised as the aforementioned injection. As in the previous proposition, the induced integral affine structure on $\mathbb{T}^2$ via pullback will have holonomy representation $\mathrm{hol} \circ \iota$ where ι is the injection of fundamental groups and hol is the holonomy representation of the affine structure on $\mathbb{T}^2$ - indeed this will be equivariant with respect to the same developing map on their shared universal cover. This shows the following proposition

Proposition 3.18. Let K^2 be equipped with an integral affine structure with holonomy representation

$$\text{hol} : \Gamma \to \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$$

Then if $\text{hol}(a) = g \in \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ and $\text{hol}(b) = h \in \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ then the induced holonomy representation on the torus $\mathbb{T}^2$ will be given by

$$\tilde{\text{hol}} : \pi_1(\mathbb{T})^2 = \langle x, y \rangle \to \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$$

$$x \mapsto g, \quad y \mapsto h^2$$

Remark 3.19. Namely, this implies that once we classify the injective homomorphisms from $\Gamma \to \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$, we can easily check which integral affine structure they induce on the torus.

3.3.2 Determining the linear holonomy

We now start are classification of the injective group morphisms from $\Gamma \to \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$. We will denote also by $a = (A, \bar{x})$ and $b = (B, \bar{y})$ the images of our generators of Γ under the holonomy representation where $A, B \in GL(2, \mathbb{Z})$ and $\bar{x}, \bar{y} \in \mathbb{R}$.

We will make use of the following in our classification

1. We must have that $\det A = 1$ and $\det B^2 = 1$ as these are generators form $\pi_1(\mathbb{T}^2)$.
2. Since b reverses orientation we in fact have $\det B = -1$
3. The action given by the composition $\pi_1(\mathbb{T}^2) \hookrightarrow \Gamma \to \text{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ must also act freely and properly as this defines the induced integral affine structure on $\mathbb{T}^2$ by Proposition 3.18.

We will now make use of Lemma 3.9 to get restrictions on the matrix B. Namely, this lemma implies that Tr $B = 0$. By Lemma 1 of [FS83], our matrix B must be conjugate to one of

$$B_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \text{ or } B_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Now we must find the matrices $A \in GL(2, \mathbb{Z})$ such that $\det A = 1$, Tr $A = 2$ which also satisfy the group relation $ABA = B$ where $B = B_1, B_2$. We perform this calculation now.

Proposition 3.20. Let $B = B_1$. Then we have the following choices for our matrix A.

1. $A = I$
2. $A_1 = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$
3. $A_2 = \begin{pmatrix} 1 & 0 \\ p & 1 \end{pmatrix}$

Proof. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL(2, \mathbb{Z})$. The trace condition implies that $d = 2 - a$. We note that $B^{-1} = B$. Calculating the group condition $ABA = B$ we obtain

$$\begin{aligned} ABA &= \begin{pmatrix} a^2 - bc & ab - bd \\ ca - cd & cb - d^2 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{aligned}$$

Comparing matrix entries we find

$$a = d$$

since $d = 2 - a$ this implies $a = d = 1$. Our matrix A now has the form

$$A = \begin{pmatrix} 1 & b \\ c & 1 \end{pmatrix}$$

From $\det A = 1$ we find that either $b = 0$ or $c = 0$ which completes the proof. ☐

Now we perform the same calculation for $B = B_2$.

Proposition 3.21. Let $B = B_2$. Then we have the following choices for our matrix A.

1. $A = I$
2. $A_3 = \begin{pmatrix} 1+n & n \\ -n & 1-n \end{pmatrix}$
3. $A_4 = \begin{pmatrix} 1-n & n \\ -n & 1+n \end{pmatrix}$

Proof. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL(2, \mathbb{Z})$. Again we m ust have that $d = 2 - a$ by the trace condition. Calculating the group law $ABA = B$ yields

$$\begin{aligned} ABA &= \begin{pmatrix} ab + ac & b^2 + ad \\ ad + c^2 & bd + cd \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{aligned}$$

Comparing entries yields that $a(b + c) = 0$ and $d(b + c) = 0$. Hoiwever, the trace condition implies that a and d can't both be zero and so $b = -c$. Our matrix A now has the form

$$A = \begin{pmatrix} a & b \\ -b & d \end{pmatrix}$$

Calculating the determinant condition and using $d = 2 - a$ yields

$$\begin{aligned} b^2 &= 1 - ad \\ &= (a-1)^2 \end{aligned}$$

and so

$$b = a - 1$$

or

$$b = 1 - a$$

The first case yields

$$A = \begin{pmatrix} n+1 & n \\ -n & 1-n \end{pmatrix}$$

where $n = a - 1$ and the second case yields

$$A = \begin{pmatrix} 1-n & n \\ -n & 1+n \end{pmatrix}$$

where $n = 1 - a$. The case where $n = 0$ corresponds to the identity and this completes our proof. □

This gives the possible choices of A and B that satisfy the group law and Lemma 3.9 we must now deduce the translational component in each case and check that the induced action is free and proper.

3.3.3 Translational component and freeness of the action

To find the translational components we impose the group law $aba = b$. When checking freeness, we can often use Proposition 3.18 to take shortcuts as we know that a, b^2 must generate one of the integral affine structures on the torus and hence act freely.

We will now compute the translational components and check whether the action is free.

Lemma 3.22. Let $B = B_1$. Then the possible choices of A that induce an integral affine structure on K^2 are $A = I, A_1$

Proof.

Case 1 ($A = I$, $B = B_1$)**.** Let $a = (I, \bar{x})$ and $b = (B_1, \bar{b})$ with $\bar{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and $\bar{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$ be our generators in $\mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$. They must satisfy the relation $aba = b$. Calculating this we find

$$(B, B\bar{x} + \bar{x} + \bar{b}) = (B, \bar{b})$$

The translational component gives $B\bar{x} = -\bar{x}$ and so $\bar{x}$ lives in the -1-eigenspace. This implies that $x_1 = 0$. We are free to conjugate both our generators as this preserves the integral affine structure and so we will do so by $g = \left(I, \begin{pmatrix} 0 \\ \frac{y_2}{2} \end{pmatrix}\right)$ Conjugating a leaves it unchanged so we calculate

$$g^{-1} \circ b \circ g = \left(B, \begin{pmatrix} y_1 \\ 0 \end{pmatrix}\right)$$

and so we may assume that $y_2 = 0$ by performing this conjugation. Our generators now look like

$$a = \left(I, \begin{pmatrix} 0 \\ x_2 \end{pmatrix}\right), \quad b = \left(B, \begin{pmatrix} y_1 \\ 0 \end{pmatrix}\right)$$

Since B alone clearly does not act freely we can't have that $y_1 = 0$ and if $x_2 = 0$ then our lattice is one-dimensional and hence the holonomy representation is not injective. Therefore, $x_2, y_1 \neq 0$. Furthermore, by changing generators $a \mapsto a^{-1}$ and $b \mapsto b^{-1}$ we can assume $x_2, y_1 > 0$. It remains to check that this action is free. We must check that $a^q b^n$ acts freely for all $q, n \in \mathbb{Z}$

not both zero. However, we now note that as $B^{2k} = I$ for $k \in \mathbb{Z}$, by the torus classification of Theorem 3.15, we have already shown that $a^q b^{2k}$ acts freely. We therefore must check the case of $a^q b^{2k+1}$. We compute that

$$a^q = \left(I, \begin{pmatrix} 0 \\ qx_2 \end{pmatrix}\right), \quad b^{2k+1} = \left(B, \begin{pmatrix} (2k+1)y_1 \\ 0 \end{pmatrix}\right)$$

For the action to be free we need $a^q b^{2k+1} \circ z = z$ to have no solutions for all $z \in \mathbb{R}^2$. We compute this action

$$\begin{aligned} a^q b^{2k+1} \circ z &= B \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} + \begin{pmatrix} (2k+1)y_1 \\ x_2 \end{pmatrix} \\ &= \begin{pmatrix} (2k+1)y_1 + z_1 \\ x_2 - z_2 \end{pmatrix} \end{aligned}$$

and so, for there to be a fixed point we need

$$(2k+1)y_1 + z_1 = z_1$$

which gives $(2k+1)y_1 = 0$. But now $y_1 > 0$ and $2k+1 = 0 \Rightarrow k = -\frac{1}{2} \notin \mathbb{Z}$ and so this action can have no fixed points. The induced integral affine structure on the torus is given by

$$a = \left(1, \begin{pmatrix} 0 \\ x_2 \end{pmatrix}\right), \quad b^2 = \left(1, \begin{pmatrix} 2y_1 \\ 0 \end{pmatrix}\right)$$

Case 2 ($A = A_1$, $B = B_1$)**.** Let $a = \left(A_1, \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right)$ and $b = \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right)$ As $A_1^2 \neq I$ and $B^2 = I$, by the torus classification we again obtain that $x_1 = 0$. We calculate the group law $aba = b$. This gives us

$$\begin{aligned} aba &= \left(B, \begin{pmatrix} y_1 + n(y_2 - x_2) \\ y_2 \end{pmatrix}\right) \\ &= \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) \end{aligned}$$

which imposes the restriction $n(y_2 - x_2) = 0$ and since $n \neq 0$ this implies $y_2 = x_2$. We can also assume that $x_2 > 0$ by conjugating the generators by $(_I, \bar{0})$ and that $y_1 > 0$ by changing generators $b \mapsto b^{-1}$ if necessary. Now we check freeness of the action.

We calculate that

$$a^q = \left(\begin{pmatrix} 1 & qn \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} \frac{nx_2 q(q-1)}{2} \\ qx_2 \end{pmatrix}\right)$$

and

$$b^k = \begin{cases} \left(B, \begin{pmatrix} ky_1 \\ x_2 \end{pmatrix}\right), \text{ k odd} \\ \left(I, \begin{pmatrix} ky_1 \\ 0 \end{pmatrix}\right), \text{ k even} \end{cases}$$

Since the case of k being even is covered by the torus classification we must only check freeness for k being odd.

For k odd we compute

$$a^q b^k = \left(\begin{pmatrix} 1 & -qn \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} \frac{nx_2 q(q-1)}{2} + qnx_2 + ky_1 \\ (q+1)x_2 \end{pmatrix}\right)$$

calculating this action on $z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$ yields

$$(a^q b^k) \circ z = \begin{pmatrix} \dfrac{nx_2 q(q-1)}{2} + qnx_2 + ky_1 + z_1 - qnz_2 \\ (q+1)x_2 - z_2 \end{pmatrix}$$

Solving the fixed point equation, the bottom entry of the vector yields

$$z_2 = \frac{q+1}{2} x_2 \tag{3.4}$$

Solving the fixed point equation in the top component yields

$$z_2 = \frac{x_2(q-1)}{2} x_2 + \frac{ky_1}{qn}$$

Substituting Equation 3.4 for z_2 yields $ky_1 = 0$ which implies $k = 0$ as $y_1 > 0$. This now implies that a^q would need to have a fixed point but one checks

$$a^q \circ z = \begin{pmatrix} * \\ z_2 + qx_2 \end{pmatrix}$$

and so we require that $qx_2 = 0$ but $x_2 > 0$ and hence the only element with a fixed point occurs when $k = q = 0$ which is the identity and hence the action is free.

The induced integral affine structure on $\mathbb{T}^2$ is given by

$$a = \left(A_1, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b^2 = \left(I, \begin{pmatrix} 2y_1 \\ 0 \end{pmatrix} \right)$$

with $x_2, y_1 > 0$

Case 3 ($A = A_2$, $B = B_1$)**.** Let $a = \left(A_1, \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right)$ and $b = \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right)$. In this case the action will fail to be free. Again, by considering the induced structure on the torus it follows that $x_1 = 0$. We calculate aba again which yields

$$\begin{aligned} aba &= \left(B, \begin{pmatrix} y_1 \\ py_1 + y_2 \end{pmatrix} \right) \\ &= \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right) \end{aligned}$$

and so we must have $py_1 = 0$ and since $p > 0$ we have $y_1 = 0$. But now we calculate b^2 as

$$b^2 = \left(I, \begin{pmatrix} 2y_1 \\ 0 \end{pmatrix} \right)$$

but since $y_1 = 0$ this cannot induce an integral affine structure on the torus hence the action is not free.

□

This completes the case where $B = B_1$. Next, we tackle the case of $B = B_2$.

Lemma 3.23. Let $B = B_2$. Then the possible choices of A that induce an integral affine structure on K^2 are $A = I, A_4$.

Proof.

Case 4 ($A = I$, $B = B_2$)**.** We calculate the group law $aba = b$ which yields

$$\begin{aligned} aba &= \left(B, \begin{pmatrix} x_1 + x_2 + y_1 \\ x_2 + y_2 + x_1 \end{pmatrix}\right) \\ &= \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) \end{aligned}$$

This implies that $x_1 = -x_2$. We conjugate a, b by

$$c = \left(-I, \begin{pmatrix} -\frac{y_2}{2} \\ -\frac{y_2}{2} \end{pmatrix}\right)$$

which gives use the new generators

$$a = \left(I, \begin{pmatrix} -x_1 \\ x_1 \end{pmatrix}\right), \quad b = \left(B, \begin{pmatrix} -y_2 - y_1 \\ 0 \end{pmatrix}\right)$$

and by switching generators $a \mapsto a^{-1}$ if necessary we may assume that $x_1 < 0$ so that we will switch the signs on the translational component of a. We will do the same for the translational component of b and assume $y_2 = 0$ and $y_1 > 0$ after relabelling $-y_2 - y_1 \to y_1$. This finishes our generating set which we write as

$$a = \left(I, \begin{pmatrix} x_1 \\ -x_1 \end{pmatrix}\right), \quad b = \left(B, \begin{pmatrix} y_1 \\ 0 \end{pmatrix}\right)$$

with $x_1, y_1 > 0$. We now check freeness of the action. Again since $B^{2k} = I$, the even case is covered by the torus classification and so we will check freeness of words $a^q b^k$ with k being odd. We calculate that

$$\begin{aligned} a^q &= \left(I, \begin{pmatrix} qx_1 \\ -qx_1 \end{pmatrix}\right) \\ b^k &= \left(B, \begin{pmatrix} \frac{k+1}{2} y_1 \\ \frac{k-1}{2} y_1 \end{pmatrix}\right), \text{ for k odd} \\ a^q b^k &= \left(B, \begin{pmatrix} \frac{k+1}{2} y_1 + qx_1 \\ \frac{k-1}{2} y_1 - qx_1 \end{pmatrix}\right), \text{ for k odd.} \end{aligned}$$

Now let $z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \in \mathbb{R}^2$ and consider the fixed point equation $(a^q b^k) \circ z = z$. This gives

$$\begin{pmatrix} \frac{k+1}{2} y_1 + qx_1 + z_2 \\ \frac{k-1}{2} y_1 - qx_1 + z_1 \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$$

Solving this gives $\frac{k+1}{2} y_1 + \frac{k-1}{2} y_1 = 0 \Rightarrow k y_1 = 0$. But since $y_1 > 0$ we must have $k = 0$. So if the action is not free we must have that a^q acts non-freely for some $q \neq 0$ but this action is by translations and hence is free and so $A = I, B = B_2$ does induce an integral affine structure. The induced structure on the torus is given by

$$a = \left(I, \begin{pmatrix} x_1 \\ -x_1 \end{pmatrix}\right), \quad b^2 = \left(I, \begin{pmatrix} y_1 \\ y_1 \end{pmatrix}\right)$$

$x_1, y_1 > 0$.

Case 5 ($A = A_3$, $B = B_2$)**.** We show that this cannot induce an integral affine structure on the torus. We conjugate A and B by

$$G = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

which puts them in the form

$$A = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}$$

and so we have $B^2 = I$. Hence, the integral affine structure on $\mathbb{T}^2$ should belong to the second family. However calculating

$$b^2 = \left(I, \begin{pmatrix} y_2 \\ 2y_2 \end{pmatrix} \right)$$

we require that $2y_2 = 0 \Rightarrow y_2 = 0$ but then b^2 does not act freely and hence does not induce an integral affine structure on the torus. Therefore, this case cannot induce an integral affine structure on K^2.

Case 6 ($A = A_4$, $B = B_2$)**.** Again, we conjugate A, B by the matrix

$$G = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$$

and assume that

$$a = \left(A = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right), \quad b = \left(B = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right)$$

Using that $B^2 = I$ and a, b^2 must induce an integral affine structure on the torus it follows that $x_1 = 0$ since the induced structure belongs to the second family. Now, calculating the group law $aba = b$ yields

$$\left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 + ny_2 + x_2(1-n) \\ y_2 \end{pmatrix} \right) = \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right)$$

and so $ny_2 + (1-n)x_2 = 0 \Rightarrow y_2 = \frac{n-1}{n}x_2$. Our generators are therefore

$$a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b = \left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ \frac{n-1}{n}x_2 \end{pmatrix} \right)$$

and the induced structure on the torus is

$$a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b^2 = \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 2y_1 + \frac{n-1}{n}x_2 \\ 0 \end{pmatrix} \right)$$

with $y_1 > 0$ and so a necessary condition is $2y_1 \neq -\frac{n-1}{n}x_2$. In fact, we shall now show that this is sufficient. We calculate that

$$a^k = \left(\begin{pmatrix} 1 & kn \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} \frac{nk(k-1)}{2}x_2 \\ kx_2 \end{pmatrix} \right), \quad b^q = \left(B, \begin{pmatrix} qy_1 + \frac{q-1}{2}y_2 \\ y_2 \end{pmatrix} \right), \text{ for q odd}$$

and

$$a^k b^q = \left(\begin{pmatrix} 1 & 1-kn \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} qy_1 + \frac{q-1}{2}y_2 + kny_2 + \frac{nk(k-1)}{2}x_2 \\ y_2 + kx_2 \end{pmatrix} \right)$$

Letting $v_1 := qy_1 + \frac{q-1}{2}y_2 + kny_2 + \frac{nk(k-1)}{2}x_2$ and $v_2 := y_2 + kx_2$ and $z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \in \mathbb{R}^2$, the fixed point equation yields

$$\begin{pmatrix} (1-kn)z_2 \\ -2z_2 \end{pmatrix} = -\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

This yields the equation $(1-kn)v_2 + 2v_1 = 0$. Solving this equation and using that $y_2 = \frac{n-1}{n}x_2$ we obtain

$$q(2y_1 + y_2) = 0$$

It is quick to check that a^k acts freely so if the action is not free then $q \neq 0$, and this gives the requirement

$$2y_1 \neq -\frac{n-1}{n}x_2$$

and so this condition is sufficient and necessary.

□

This completes the classification of the integral affine structures on the torus. We summarize the results of the previous two lemmas into the following theorem.

Theorem 3.24. The possible integral integral affine structures on the Klein bottle are

1. $a = \left(I, \begin{pmatrix} 0 \\ x_2 \end{pmatrix}\right), \quad b = \left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ 0 \end{pmatrix}\right)$, with $x_2, y_1 > 0$.
2. $a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix}\right), \quad b = \left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ x_2 \end{pmatrix}\right)$, with $x_2, y_1 > 0$.
3. $a = \left(I, \begin{pmatrix} x_1 \\ -x_1 \end{pmatrix}\right), \quad b = \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} y_1 \\ 0 \end{pmatrix}\right)$, with $x_1, y_1 > 0$.
4. $a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix}\right), \quad b = \left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ \frac{n-1}{n}x_2 \end{pmatrix}\right)$, with $y_1 > 0$ and $2y_1 \neq -\frac{n-1}{n}x_2$.

We note that families 1) and 3) induce a Euclidean structure on the Klein bottle, e.g. the preserves some Riemannian metric as we have that they group they generate is a subset of $O(n) \rtimes \mathbb{R}^2$.

We now wish to classify the corresponding Lagrangian fibrations for each of the integral affine structures on the two torus and the Klein bottle. Below are some pictures of some of the different families of integral affine structures on both the torus and the Klein bottle. (TODO: generate the lattices)

CHAPTER FOUR

Classification of Lagrangian Fibrations over Closed Surfaces

Conclusion

TODO: Add Conclusion

Bibliography

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